

trace of A . In particular, Exercises 37 and 38 show that the stability of the origin can be described as in Figure 8.

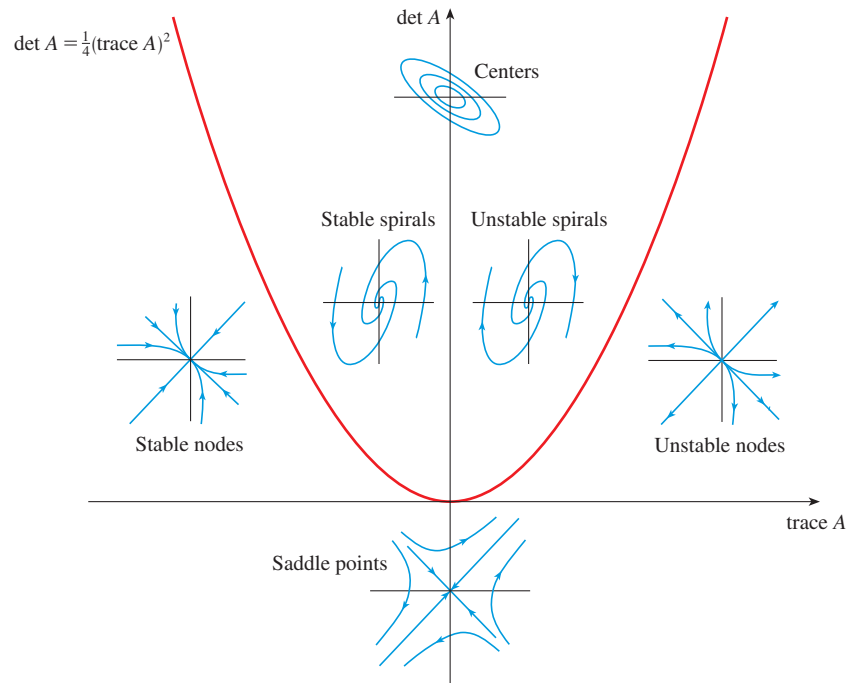


FIGURE 8

The stability properties of the equilibrium at the origin in terms of the determinant and trace of A .

EXERCISES 10.2

1–4 Show that $x_1(t)$ and $x_2(t)$ are solutions to the system of differential equations $d\mathbf{x}/dt = A\mathbf{x}$.

1. $A = \begin{bmatrix} 3 & -2 \\ 2 & -2 \end{bmatrix}$

$$x_1(t) = \frac{1}{3}(4e^{2t} - e^{-t}), \quad x_2(t) = \frac{2}{3}(e^{2t} - e^{-t})$$

2. $A = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$

$$x_1(t) = \frac{1}{2}(e^{-3t} + e^{-t}), \quad x_2(t) = \frac{1}{2}(-e^{-3t} + e^{-t})$$

3. $A = \begin{bmatrix} -1 & -4 \\ 1 & -1 \end{bmatrix}$

$$x_1(t) = e^{-t} \cos 2t, \quad x_2(t) = \frac{1}{2}e^{-t} \sin 2t$$

4. $A = \begin{bmatrix} 1 & 2 \\ -4 & 1 \end{bmatrix}$

$$x_1(t) = e^t \cos(2\sqrt{2}t), \quad x_2(t) = -\sqrt{2}e^t \sin(2\sqrt{2}t)$$

5–8 Show that $x_1(t)$ and $x_2(t)$ are solutions to the initial-value problem $d\mathbf{x}/dt = A\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$.

5. $A = \begin{bmatrix} 3 & -2 \\ 6 & -4 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$x_1(t) = e^{-t}, \quad x_2(t) = 2e^{-t}$$

6. $A = \begin{bmatrix} -2 & 1 \\ -5 & 4 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

$$x_1(t) = e^{3t}, \quad x_2(t) = 5e^{3t}$$

7. $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$x_1(t) = e^t(2 \cos t - \sin t), \quad x_2(t) = e^t(\cos t + 2 \sin t)$$

8. $A = \begin{bmatrix} -3 & 2 \\ -1 & -1 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$$x_1(t) = e^{-2t}(-2 \cos t + 4 \sin t),$$

$$x_2(t) = e^{-2t}(\cos t + 3 \sin t)$$

9. Prove the Superposition Principle.
10. Show that if the eigenvalues of a 2×2 matrix are real and distinct, then the matrix P whose columns are the corresponding eigenvectors is nonsingular.

11–16 Sketch several solution curves in the phase plane of the system of differential equations $d\mathbf{x}/dt = A\mathbf{x}$ using the given eigenvalues and eigenvectors of A .

11. $\lambda_1 = -1, \lambda_2 = -2; \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

12. $\lambda_1 = 2, \lambda_2 = 4; \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

13. $\lambda_1 = 2, \lambda_2 = -2; \quad \mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

14. $\lambda_1 = -3, \lambda_2 = -2; \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

15. $\lambda_1 = 5, \lambda_2 = 1; \quad \mathbf{v}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

16. $\lambda_1 = 1, \lambda_2 = -1; \quad \mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$

17–28 Solve the initial value problem $d\mathbf{x}/dt = A\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$.

17. $A = \begin{bmatrix} -\frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{3}{2} \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

18. $A = \begin{bmatrix} \frac{1}{2} & -\frac{3}{2} \\ -\frac{3}{2} & \frac{1}{2} \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

19. $A = \begin{bmatrix} 1 & 0 \\ 4 & -1 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

20. $A = \begin{bmatrix} -1 & -2 \\ 2 & -2 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

21. $A = \begin{bmatrix} -3 & 4 \\ -6 & 7 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

22. $A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$

23. $A = \begin{bmatrix} -1 & 2 \\ -3 & -1 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

24. $A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$

25. $A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

26. $A = \begin{bmatrix} 4 & -2 \\ 3 & 1 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

27. $A = \begin{bmatrix} 2 & -5 \\ 2 & 1 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

28. $A = \begin{bmatrix} 3 & -4 \\ 1 & -3 \end{bmatrix} \quad \mathbf{x}_0 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

29. In Exercise 10.1.24 we considered the *nongeneric* system of differential equations

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} -2 & -1 \\ 2 & 1 \end{bmatrix} \mathbf{x}$$

Theorem 2 applies to this system, and we can obtain the general solution (8) in the usual way. Do so.

30. When the eigenvalues of the coefficient matrix are complex, the origin is a spiral. If the eigenvalues are purely imaginary (that is, the real parts are zero), then the origin is called a *center*. For example, this is true of the following system:

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} -1 & 2 \\ -3 & 1 \end{bmatrix} \mathbf{x}$$

Theorem 2 still applies to this system, and we can obtain the general solution (13) in the usual way.

- (a) Find the general solution.
 (b) Use the general solution in part (a) to prove that the solutions form closed curves in the phase plane.

Repeated Eigenvalues

31. Our focus has been on systems whose coefficient matrices have distinct eigenvalues. A simple example of a system with **repeated eigenvalues** is

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$

- (a) Show that $x_1(t) = c_1 e^{-t}$ and $x_2(t) = c_2 e^{-t}$ is a solution. The origin in this case is called a **proper node**.
 (b) Try obtaining this general solution by calculating the eigenvectors and eigenvalues of the coefficient matrix. Comment on anything unusual that occurs.

32. A slightly more complicated system with **repeated eigenvalues** is

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$

- (a) Show that $x_1(t) = c_1 e^{-t} + c_2 t e^{-t}$ and $x_2(t) = c_2 e^{-t}$ is a solution. The origin in this case is called an **improper node**.
 (b) Try obtaining this general solution by calculating the eigenvectors and eigenvalues of the coefficient matrix. Comment on anything unusual that occurs.

33–35 The system of differential equations $d\mathbf{x}/dt = A\mathbf{x}$ depends on a real-valued constant a . Use the eigenvalues to determine the stability properties of the equilibrium at the origin for all values of a .

33. $A = \begin{bmatrix} -1 & 1 \\ a & -1 \end{bmatrix}$

34. $A = \begin{bmatrix} -1 & a \\ 1 & 1 \end{bmatrix}$

35. $A = \begin{bmatrix} a & 1 \\ 1 & a \end{bmatrix}$

36. Use the general solution (Equation 8) and Euler's formula to prove Theorem 15 for the case where the eigenvalues of the coefficient matrix A are distinct.
37. Use Theorem 15 to prove the **trace and determinant condition for stability** given by Theorem 16. [Hint: Express the equations for the eigenvalues of a 2×2 matrix

in terms of the trace and determinant, and then use Theorem 15].

38. Justify the summary of the qualitative behavior depicted in Figure 8. Be sure to explain where the curve defined by $\det A = \frac{1}{4}(\text{trace } A)^2$ comes from.
39. Provide an argument, based on Theorem 2, for why solution curves in the phase plane of a two-dimensional, autonomous system of linear differential equations cannot cross.

10.3 Applications

We now illustrate how the results from the preceding sections can be applied to a variety of areas in the life sciences.

Metapopulations

Many biological populations are subdivided into smaller subpopulations, each living in its own patch, with limited movement between them. The entire collection of such subpopulations is called a **metapopulation**. Consider the following model of two subpopulations, where x_A and x_B are the number of individuals in each:

$$\begin{aligned}\frac{dx_A}{dt} &= r_A x_A - m_A x_A + m_B x_B \\ \frac{dx_B}{dt} &= r_B x_B - m_B x_B + m_A x_A\end{aligned}$$

Here r_i is the per capita growth rate of subpopulation i , and m_i is the per capita movement rate from patch i into the other patch. A specific example of the model is shown in Figure 1, where $r_A = 1$, $r_B = -2$, $m_A = 2$, and $m_B = 0$.

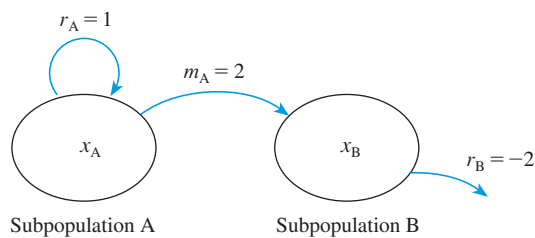


FIGURE 1

With these choices the equations become

$$\begin{aligned}\frac{dx_A}{dt} &= x_A - 2x_A \\ \frac{dx_B}{dt} &= -2x_B + 2x_A\end{aligned}$$

Individuals in subpopulation A have a net per capita growth rate of $r_A = 1$, while those in subpopulation B die off, having a net per capita growth rate of $r_B = -2$. Individuals in subpopulation A also move to subpopulation B at per capita rate $m_A = 2$, whereas individuals in subpopulation B never move (see Figure 1). We suppose the initial size of each subpopulation is $x_A(0) = 100$ and $x_B(0) = 50$.

What is the size of each subpopulation as a function of time? Answering this question requires that we solve an initial-value problem. In matrix notation we have

$$(1) \quad \frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

with

$$(2) \quad A = \begin{bmatrix} -1 & 0 \\ 2 & -2 \end{bmatrix} \quad \text{and} \quad \mathbf{x}(0) = \begin{bmatrix} 100 \\ 50 \end{bmatrix}$$

The eigenvalues of matrix A are readily found to be $\lambda_1 = -2$ and $\lambda_2 = -1$. (Recall from Exercise 8.7.17 that the eigenvalues of a lower triangular matrix lie on the diagonal.) Therefore, from the results of Section 10.2, the origin in this model is a stable node. The metapopulation therefore goes extinct as $t \rightarrow \infty$.

The eigenvectors are found to be

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Therefore the general solution to the differential equation (1) is

$$(3) \quad \mathbf{x}(t) = c_1 e^{-2t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

To find the constants c_1 and c_2 we make use of the initial condition. At $t = 0$, Equation 3 is

$$\mathbf{x}(0) = c_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Therefore we must choose c_1 and c_2 to satisfy the equation

$$\begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 100 \\ 50 \end{bmatrix}$$

We obtain $c_1 = -150$ and $c_2 = 100$. The solution to the initial-value problem is therefore

$$(4) \quad \mathbf{x}(t) = -150e^{-2t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 100e^{-t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Figure 2 shows the phase plane for this model, along with the nullclines, the eigenvectors, and the solution (4). Biologically, we are interested in only the positive quadrant of the phase plane.

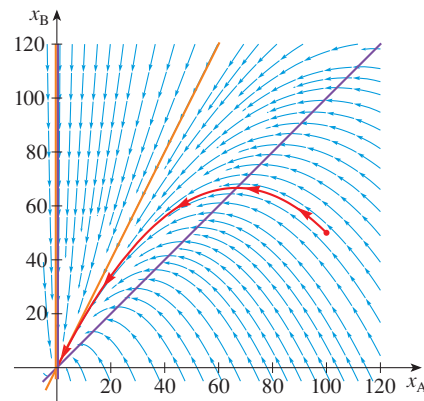


FIGURE 2

Purple lines are nullclines. Orange lines are all scalar multiples of the eigenvectors. Blue curves show a family of solutions. The red curve is the solution to the initial-value problem.

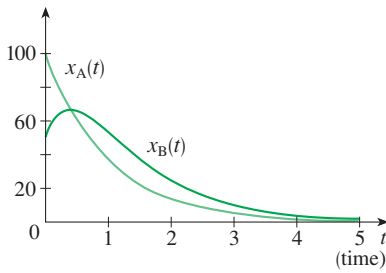


FIGURE 3

To better understand the fate of each individual subpopulation, we can look at each component of Equation 4 individually. They are

$$\begin{aligned}x_A(t) &= 100e^{-t} \\x_B(t) &= 200e^{-t} - 150e^{-2t}\end{aligned}$$

Figure 3 plots each component of the solution as a function of time. Both Figure 2 and Figure 3 reveal that the metapopulation ultimately goes extinct. Subpopulation A immediately declines to extinction, whereas subpopulation B initially increases in size before eventually going extinct.

■ Natural Killer Cells and Immunity

Natural killer cells (often abbreviated NK cells) are components of the human immune system that defend the body against a variety of diseases, including cancers and pathogens (see Figure 4). NK cells are typically found in one of two states: latent or actively dividing. Suppose that latent cells die at a per capita rate d and are recruited to the dividing state at per capita rate r . Dividing cells replicate at a per capita rate k and die at a per capita rate h . All constants are positive.

These assumptions lead to the following system of differential equations describing each kind of NK cell, where $L(t)$ and $D(t)$ are the numbers of latent and dividing cells:¹

$$\begin{aligned}\frac{dL}{dt} &= -rL - dL \\ \frac{dD}{dt} &= rL + (k - h)D\end{aligned}$$

At $t = 0$ all NK cells are in the latent state and therefore the initial condition is $L(0) = L_0$ and $D(0) = 0$. We want to determine the number of each NK cell type as a function of time.

Estimates exist for each of the constants in this model, but these values change under different conditions. We could substitute any given constant values into the model and then solve the initial-value problem, but it would be better to have a solution in terms of arbitrary constants so that we could apply our results under different conditions without repeatedly having to solve the initial-value problem. This approach is illustrated here.

If $\mathbf{x}(t)$ denotes the vector whose components are $L(t)$ and $D(t)$, the model can be written in matrix notation as

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

with

$$A = \begin{bmatrix} -(r + d) & 0 \\ r & k - h \end{bmatrix} \quad \text{and} \quad \mathbf{x}(0) = \begin{bmatrix} L_0 \\ 0 \end{bmatrix}$$

Matrix A is lower triangular and therefore its eigenvalues lie on the diagonal. They are $\lambda_1 = -(r + d)$ and $\lambda_2 = k - h$. All constants are positive and therefore $\lambda_1 < 0$, whereas λ_2 is positive if $k > h$ and negative if the reverse inequality holds. Therefore, from the results of Section 10.2, the origin is a stable node if $k < h$ and it is a saddle if $k > h$.

1. Y. Zhao et al., “Two-Compartment Model of NK Cell Proliferation: Insights from Population Response to IL-15 Stimulation,” *Journal of Immunology* 188 (2012): 2981–90.

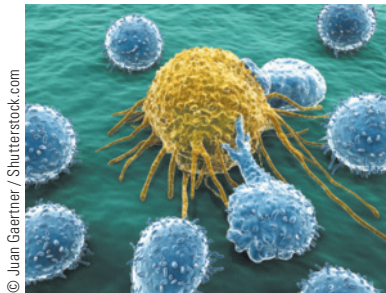


FIGURE 4
NK cells attacking a pathogen

The eigenvectors are found to be

$$\mathbf{v}_1 = \begin{bmatrix} h - d - k - r \\ r \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore the general solution is

$$(5) \quad \mathbf{x}(t) = c_1 e^{-(r+d)t} \begin{bmatrix} h - d - k - r \\ r \end{bmatrix} + c_2 e^{(k-h)t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

To obtain the unique member of this family that solves the initial-value problem, we use the initial condition. At $t = 0$, Equation 5 is

$$\mathbf{x}(0) = c_1 \begin{bmatrix} h - d - k - r \\ r \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} h - d - k - r & 0 \\ r & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Therefore c_1 and c_2 must satisfy the equation

$$\begin{bmatrix} h - d - k - r & 0 \\ r & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} L_0 \\ 0 \end{bmatrix}$$

Solving, we obtain

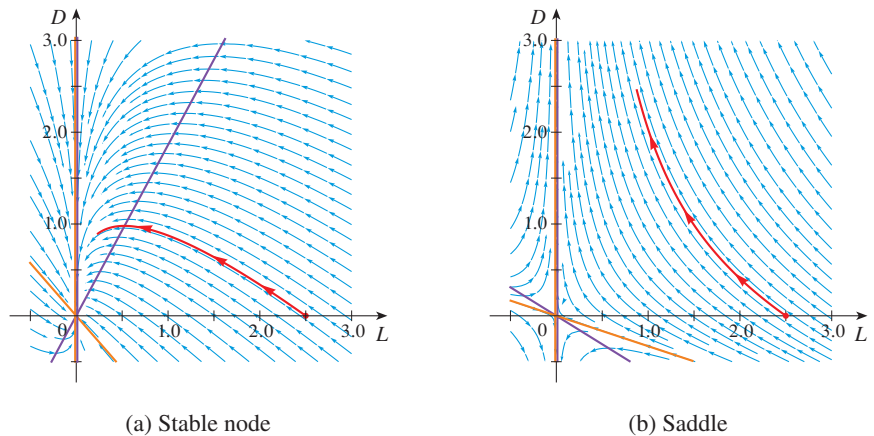
$$c_1 = \frac{L_0}{h - d - k - r} \quad \text{and} \quad c_2 = \frac{-rL_0}{h - d - k - r}$$

Therefore the solution to the initial-value problem is

$$(6) \quad \mathbf{x}(t) = L_0 e^{-(r+d)t} \begin{bmatrix} 1 \\ r \\ h - d - k - r \end{bmatrix} - L_0 e^{(k-h)t} \begin{bmatrix} 0 \\ r \\ h - d - k - r \end{bmatrix}$$

The phase plane for Equation 6 can now be constructed for any constants of interest. Figure 5 shows two examples, one in which the origin is a stable node and another in which it is a saddle.

FIGURE 5
Purple lines are nullclines. Orange lines are all scalar multiples of the eigenvectors. Blue curves show a family of solutions. The red curve is the solution to the initial-value problem.



In this example it is of primary interest to examine the total number of cells $L + D$ as a function of time. Writing each component of Equation 6 individually gives

$$L(t) = L_0 e^{-(r+d)t}$$

$$D(t) = L_0 r \frac{e^{-(r+d)t} - e^{(k-h)t}}{h - d - k - r}$$

Figure 6 plots the sum $L(t) + D(t)$ for two different sets of values for the constants, along with experimental data for each.² In both cases the constants are such that the origin is a saddle.

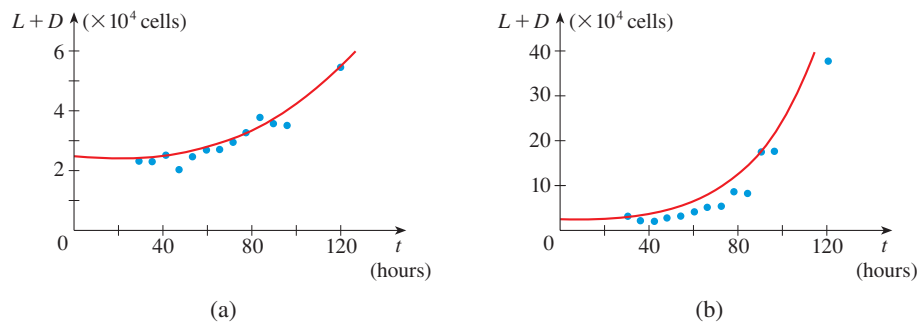


FIGURE 6



FIGURE 7

■ Gene Regulation

A gene produces protein by first creating an intermediate molecule called mRNA. High gene expression leads to a lot of mRNA molecules which, in turn, leads to a lot of protein. High levels of protein can suppress the production of mRNA, however, thereby regulating gene expression, as depicted in Figure 7.

Suppose $m(t)$ and $p(t)$ are the amounts of mRNA and protein in a cell (in hundreds of copies) at time t (in hours). A simple model of gene regulation is

$$\frac{dm}{dt} = f(p) - dm \quad \frac{dp}{dt} = am - hp$$

where d is the rate at which mRNA molecules are degraded, a is the rate at which each mRNA molecule produces protein, and h is the rate at which protein molecules are degraded. The function $f(p)$ specifies the rate of production of mRNA (in hundreds of copies per hour) as a function of the amount of protein in the cell.

In this example we suppose that $f(p) = 1 - p$, meaning that the maximal rate of mRNA production is 100 copies per hour, and it decreases to 0 as the amount of protein increases to 100 copies. Further, for simplicity we use the values $d = 1$, $a = 1$, and $h = 1$. The model then reduces to

$$\frac{dm}{dt} = 1 - p - m \quad \frac{dp}{dt} = m - p$$

or, in matrix notation,

$$(7) \quad \begin{bmatrix} \frac{dm}{dt} \\ \frac{dp}{dt} \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} m \\ p \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Suppose that, initially, there is no mRNA or protein; that is, $m(0) = 0$ and $p(0) = 0$.

Equation 7 is different from most models we have studied so far in that it is non-homogeneous. As Exercise 10.1.23 shows, however, equations like (7) can be reduced to a system of homogeneous linear differential equations with a change of variables. We first find the equilibrium $(\hat{m}, \hat{p})$ of Equation 7 and then define the new variables $x_1(t) = m(t) - \hat{m}$ and $x_2(t) = p(t) - \hat{p}$.

2. Y. Zhao et al., "Two-Compartment Model of NK Cell Proliferation: Insights from Population Response to IL-15 Stimulation," *Journal of Immunology* 188 (2012): 2981–90.

Setting both $dm/dt = 0$ and $dp/dt = 0$ gives the equation

$$\begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \hat{m} \\ \hat{p} \end{bmatrix} = -\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

We find that the equilibrium is $\hat{m} = \frac{1}{2}$ and $\hat{p} = \frac{1}{2}$. Therefore the new variables are $x_1(t) = m(t) - \frac{1}{2}$ and $x_2(t) = p(t) - \frac{1}{2}$. These represent the difference in the amounts from their equilibrium values.

With this change of variables we now proceed to model x_1 and x_2 . Following the approach of Exercise 10.1.23, we see that the variables x_1 and x_2 satisfy the initial-value problem:

$$(8) \quad \frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

with

$$A = \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} \quad \text{and} \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

The eigenvalues of A are the complex conjugates $\lambda_1 = -1 + i$ and $\lambda_2 = -1 - i$. Therefore, from the results of Section 10.2, the origin is a stable spiral. In other words, both x_1 and x_2 approach zero as $t \rightarrow \infty$. This means that the variables m and p approach their equilibrium values as $t \rightarrow \infty$.

The eigenvectors associated with each eigenvalue are

$$\mathbf{v}_1 = \begin{bmatrix} i \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

Using the notation in Expression 10.2.11, we have

$$\mathbf{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Therefore, using Equation 10.2.13, we obtain the general solution

$$(9) \quad \mathbf{x}(t) = k_1 e^{-t}(\mathbf{u} \cos t - \mathbf{w} \sin t) + k_2 e^{-t}(\mathbf{w} \cos t + \mathbf{u} \sin t)$$

To find the constants k_1 and k_2 we make use of the initial condition. At $t = 0$, Equation 9 is

$$\mathbf{x}(0) = k_1 \mathbf{u} + k_2 \mathbf{w} = k_1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$$

Therefore k_1 and k_2 must satisfy the equation

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

This equation requires that $k_1 = -\frac{1}{2}$ and $k_2 = -\frac{1}{2}$. Thus the solution to the initial-value problem for $\mathbf{x}(t)$ is

$$(10) \quad \mathbf{x}(t) = -\frac{1}{2} e^{-t}(\mathbf{u} \cos t - \mathbf{w} \sin t) - \frac{1}{2} e^{-t}(\mathbf{w} \cos t + \mathbf{u} \sin t)$$

Substituting in the definitions of $\mathbf{u}$ and $\mathbf{w}$ gives the components of the solution (10) as

$$x_1(t) = \frac{1}{2} e^{-t}(\sin t - \cos t)$$

$$x_2(t) = -\frac{1}{2} e^{-t}(\sin t + \cos t)$$

The final step is to obtain the solution to the initial-value problem for the original variables m and p . From the definitions of x_1 and x_2 we simply need to add $\frac{1}{2}$ to each component of the solutions for x_1 and x_2 , giving

$$m(t) = \frac{1}{2}e^{-t}(\sin t - \cos t) + \frac{1}{2}$$
$$p(t) = -\frac{1}{2}e^{-t}(\sin t + \cos t) + \frac{1}{2}$$

Figure 8 shows the phase plane for this model, along with the nullclines and the solution. Figure 9 plots m and p as functions of time. Figures 8 and 9 together show that the amounts of mRNA and protein in the cell oscillate slightly and both approach a nonzero equilibrium value.

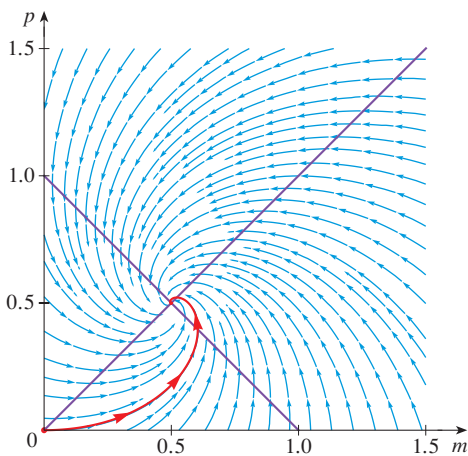


FIGURE 8
Purple lines are nullclines. Blue curves show a family of solutions. The red curve is the solution to the initial-value problem.

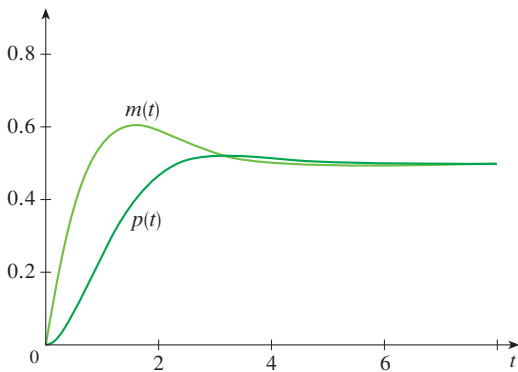


FIGURE 9
The components $m(t)$ and $p(t)$ of the red solution curve in Figure 8 are plotted against time.

■ **Transport of Environmental Pollutants**

This final example illustrates how we must sometimes manipulate a model before attempting to solve it. Suppose three lakes of equal volume are interconnected, as in Figure 10, with a net flow of water in the directions shown. A shipping accident releases 300,000 kg of a chemical pollutant into Lake 1. We would like to predict the amount of pollutant in each lake as a function of time.

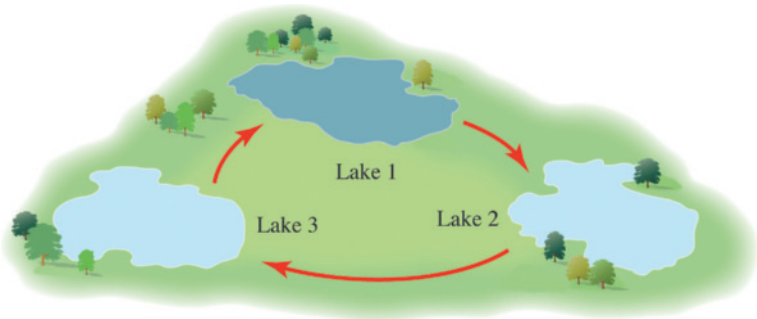


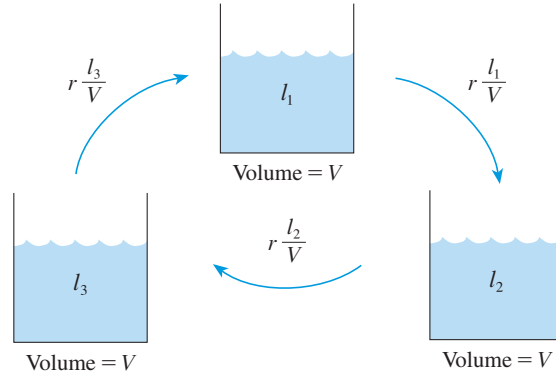
FIGURE 10

Let’s construct a simplified model in which each lake is assumed to be well mixed and none of the pollutant settles out. The variables $l_1(t)$, $l_2(t)$, and $l_3(t)$ denote the mass of pollutant in each lake at time t . Each lake is viewed as a container with volume V , and

Our model is a mixing model similar to those of Exercises 13–18 in the Review Section of this chapter.

we suppose that the rate of flow of water between any pair of lakes is r , as illustrated in Figure 11.

FIGURE 11



Each lake has an outflow and an inflow of pollutant. The outflow is the rate of flow of water r (in L/day) multiplied by the concentration of pollutant in the lake (in kg/L). The inflow is similarly the rate of flow r multiplied by the concentration of pollutant within the inflow (see Figure 11). This gives the following system of differential equations

$$\begin{aligned}\frac{dl_1}{dt} &= -r \frac{l_1}{V} + r \frac{l_3}{V} \\ \frac{dl_2}{dt} &= -r \frac{l_2}{V} + r \frac{l_1}{V} \\ \frac{dl_3}{dt} &= -r \frac{l_3}{V} + r \frac{l_2}{V}\end{aligned}$$

The initial condition is $l_1(0) = 300,000$ kg, $l_2(0) = 0$, and $l_3(0) = 0$.

Our model is a homogeneous system of linear differential equations, but it is three-dimensional, and our techniques so far have focused on systems of two equations. A bit of insight, however, allows us to reduce the system to two equations.

Since none of the pollutant ever settles out of the lakes, we would expect the total amount of pollutant in all three lakes to remain constant (at 300,000 kg). That is, we would expect the sum $l_1(t) + l_2(t) + l_3(t)$ to remain constant through time. To see that this is so, we differentiate this sum, obtaining

$$\begin{aligned}\frac{d}{dt}(l_1 + l_2 + l_3) &= \frac{dl_1}{dt} + \frac{dl_2}{dt} + \frac{dl_3}{dt} \\ &= -r \frac{l_1}{V} + r \frac{l_3}{V} - r \frac{l_2}{V} + r \frac{l_1}{V} - r \frac{l_3}{V} + r \frac{l_2}{V} \\ &= 0\end{aligned}$$

This reveals that we need only two differential equations to track the pollutant. Let's track the amount in Lakes 1 and 2, with the amount in Lake 3 being $l_3(t) = 300,000 - l_1(t) - l_2(t)$. Making this substitution in the first two differential equations of our model then gives the system

$$\begin{aligned}\frac{dl_1}{dt} &= -r \frac{l_1}{V} + r \frac{300,000 - l_1 - l_2}{V} \\ \frac{dl_2}{dt} &= -r \frac{l_2}{V} + r \frac{l_1}{V}\end{aligned}$$

where the third, redundant equation has been dropped.

Before proceeding, let's simplify the notation by defining the constant $\alpha = r/V$. In matrix notation the model is then

$$\frac{d\mathbf{l}}{dt} = \begin{bmatrix} -2\alpha & -\alpha \\ \alpha & -\alpha \end{bmatrix} \mathbf{l} + \begin{bmatrix} 300,000\alpha \\ 0 \end{bmatrix}$$

where $\mathbf{l}$ is the vector whose components are l_1 and l_2 .

We have reduced the system of equations from three to two, but it is no longer homogeneous. Therefore the next step is to change variables as we did in the preceding example. We must first find the equilibrium values, $\hat{l}_1$ and $\hat{l}_2$, and then define the new variables $x_1(t) = l_1(t) - \hat{l}_1$ and $x_2(t) = l_2(t) - \hat{l}_2$.

Setting both $dl_1/dt = 0$ and $dl_2/dt = 0$ gives the equation

$$\begin{bmatrix} -2\alpha & -\alpha \\ \alpha & -\alpha \end{bmatrix} \begin{bmatrix} \hat{l}_1 \\ \hat{l}_2 \end{bmatrix} = -\begin{bmatrix} 300,000\alpha \\ 0 \end{bmatrix}$$

from which we obtain the values $\hat{l}_1 = 10^5$ and $\hat{l}_2 = 10^5$. Therefore the new variables are $x_1(t) = l_1(t) - 10^5$ and $x_2(t) = l_2(t) - 10^5$.

With this change of variables we then obtain the following initial-value problem for x_1 and x_2 :

$$(11) \quad \frac{d\mathbf{x}}{dt} = A\mathbf{x}$$

with

$$A = \begin{bmatrix} -2\alpha & -\alpha \\ \alpha & -\alpha \end{bmatrix} \quad \text{and} \quad \mathbf{x}(0) = \begin{bmatrix} 3 \times 10^5 \\ 0 \end{bmatrix} - \begin{bmatrix} 10^5 \\ 10^5 \end{bmatrix} = \begin{bmatrix} 2 \times 10^5 \\ -10^5 \end{bmatrix}$$

The eigenvalues of A are the complex conjugates

$$\lambda_1 = -\frac{3}{2}\alpha + i\frac{\sqrt{3}}{2}\alpha \quad \lambda_2 = -\frac{3}{2}\alpha - i\frac{\sqrt{3}}{2}\alpha$$

Therefore the origin is a stable spiral because the real parts of the eigenvalues are negative. In other words, both x_1 and x_2 approach zero as $t \rightarrow \infty$. This means that the original variables l_1 and l_2 approach their equilibrium values as $t \rightarrow \infty$.

The eigenvectors associated with each eigenvalue are

$$\mathbf{v}_1 = \begin{bmatrix} -\frac{1}{2} + \frac{\sqrt{3}}{2}i \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} -\frac{1}{2} - \frac{\sqrt{3}}{2}i \\ 1 \end{bmatrix}$$

Using the notation in Expression 10.2.11, we have

$$\mathbf{u} = \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ 0 \end{bmatrix}$$

Therefore, using Equation 10.2.13, we obtain the general solution

$$(12) \quad \begin{aligned} \mathbf{x}(t) = & k_1 e^{-(3/2)\alpha t} \left(\mathbf{u} \cos \frac{\sqrt{3}}{2} \alpha t - \mathbf{w} \sin \frac{\sqrt{3}}{2} \alpha t \right) \\ & + k_2 e^{-(3/2)\alpha t} \left(\mathbf{w} \cos \frac{\sqrt{3}}{2} \alpha t + \mathbf{u} \sin \frac{\sqrt{3}}{2} \alpha t \right) \end{aligned}$$

At $t = 0$, Equation 12 reduces to

$$\mathbf{x}(0) = k_1 \mathbf{u} + k_2 \mathbf{w} = k_1 \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} \frac{\sqrt{3}}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$$

Therefore k_1 and k_2 must satisfy the equation

$$\begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 2 \times 10^5 \\ -10^5 \end{bmatrix}$$

Using Theorem 8.6.4, we therefore obtain $k_1 = -10^5$ and $k_2 = \sqrt{3} \times 10^5$.

The final step is to obtain the solution to the initial-value problem for the original variables l_1 and l_2 using (12). From the definitions of x_1 and x_2 we simply need to add 10^5 to each component of the vector given in (12). Figure 12 shows the phase plane for this model, along with the nullclines and the solution when $\alpha = 0.2$. Figure 13 plots l_1 , l_2 , and l_3 as functions of time. Figure 12 and Figure 13 both show that the amount of pollutant in each lake displays a slight oscillatory behavior. All lakes eventually contain the same amount of pollutant as t gets large.

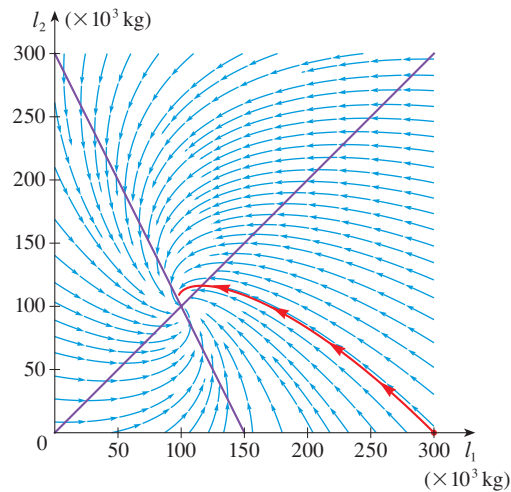


FIGURE 12

Purple lines are nullclines. Blue curves show a family of solutions. The red curve is the solution to the initial-value problem.

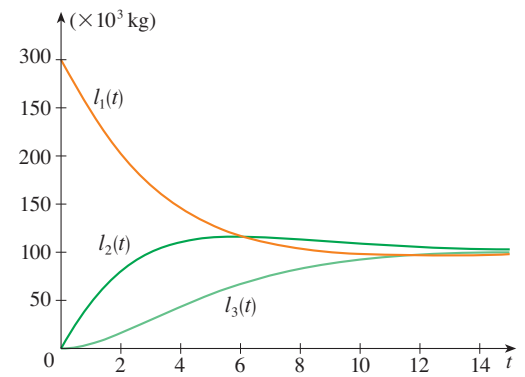


FIGURE 13

The components l_1 and l_2 of the red solution curve in Figure 12 are plotted against time, along with $l_3 = 300,000 - l_1 - l_2$.

EXERCISES 10.3

- Jellyfish locomotion** In Exercise 10.1.34 we introduced a model of jellyfish locomotion. It can be written as the following system of two linear differential equations

$$z'_1 = z_2 \quad z'_2 = -\frac{k}{m} z_1 - \frac{b}{m} z_2$$

where z_1 is the displacement of the bell at time t and z_2 is its velocity. Find the general solution when $m = 100$ g, $b = 0.1$ N/m·s, and $k = 1$ N/m.

- Hemodialysis** is a process by which a machine is used to filter urea and other waste products from a patient's blood if

the kidneys fail. The amount of urea within a patient during dialysis is sometimes modeled by supposing there are two compartments within the patient: the blood, which is directly filtered by the dialysis machine, and another compartment that cannot be directly filtered but that is connected to the blood. A system of two differential equations describing this is

$$\frac{dc}{dt} = -\frac{K}{V} c + ap - bc \quad \frac{dp}{dt} = -ap + bc$$

where c and p are the urea concentrations in the blood and

the inaccessible pool (in mg/mL) and all constants are positive (see also Exercise 14 in the Review Section of this chapter). Suppose that $K/V = 1$, $a = b = \frac{1}{2}$, and the initial urea concentration is $c(0) = c_0$ and $p(0) = c_0$ mg/mL.

- Classify the equilibrium of this system.
- Solve this initial-value problem.

- 3. Prostate cancer treatment** During the treatment of prostate cancer some tumor cells become resistant to medication through a variety of biochemical changes. Some of these changes are reversible and some are irreversible. In Exercise 10.1.32 we introduced a three-variable model for the different cell types. Under certain assumptions this can be reduced to the following two differential equations:

$$y_1' = -4y_1 \quad y_2' = y_1 - 2y_2 + d$$

where y_1 and y_2 are the fractions of cells that are sensitive and irreversibly resistant to treatment, respectively.

- Use a change of variable to reduce this system to a homogeneous system of linear differential equations.
- Find the general solution to the system from part (a).
- Suppose that, at the beginning of treatment, all cells are sensitive [that is, $y_1(0) = 1$ and $y_2(0) = 0$]. What is the solution to this initial-value problem?
- What is the function specifying the fraction of cells that are irreversibly resistant as a function of time?
- How long after the start of treatment will 50% of the cells be irreversibly resistant? (Assume that $d > 1$.)

Source: Adapted from Y. Hirata et al., "Development of a Mathematical Model that Predicts the Outcome of Hormone Therapy for Prostate Cancer," *Journal of Theoretical Biology* 264 (2010): 517–27.

- 4. Soil contamination** A crop is planted in soil that is contaminated with a pollutant. The pollutant gradually leaches out of the soil but is also absorbed by the growing crop. A simple model of this process is

$$\frac{ds}{dt} = -\alpha s - \beta s \quad \frac{dc}{dt} = \beta s$$

where s and c are the amounts of pollutant in the soil and crop (in mg), respectively, and α and β are positive constants.

- Suppose that $s(0) = s_0$ and $c(0) = 0$. What is the solution to the initial-value problem?
- In the long term (that is, as $t \rightarrow \infty$), what is the amount of pollutant in the crop?

- 5. Metastasis of malignant tumors** Metastasis is the process by which cancer cells spread throughout the body and initiate tumors in various organs. This sometimes happens via the bloodstream, where cancer cells become lodged in capillaries of organs and then move across the capillary wall into the organ. Using C to denote the number of cells lodged in a capillary and I for the number that have invaded the organ, we can model this as

$$C' = -\alpha C - \beta C \quad I' = \alpha C - \delta I + \rho I$$

where all constants are positive, α is the rate of movement across the capillary wall, β is the rate of dislodgment from

the capillary, δ is the rate at which cancer cells in the organ die, and ρ is their growth rate.

- Find the general solution.
- Classify the equilibrium at the origin when $\rho > \delta$ and when $\rho < \delta$.
- What is the solution to the initial-value problem if $C(0) = C_0$ and $I(0) = 0$?
- Use your result from part (c) to show that the tumor will grow in the long term if and only if $\rho > \delta$.

Source: Adapted from D. Kaplan et al., *Understanding Nonlinear Dynamics* (New York: Springer-Verlag, 1995).

- 6. Radioimmunotherapy** is a cancer treatment in which radioactive atoms are attached to tumor-specific antibody molecules and then injected into the blood. The antibody molecules then attach only to tumor cells, where they then deliver the cell-killing radioactivity. The following model for this process was introduced in Example 10.1.1:

$$\frac{dx_1}{dt} = -ax_1 - bx_1 \quad \frac{dx_2}{dt} = bx_1 - cx_2$$

where x_1 denotes the amount of antibody in the blood and x_2 the amount of antibody taken up by the tumor (both in μg). All constants are positive.

- Find the general solution.
- Suppose that $x_1(0) = x_0$ and $x_2(0) = 0$. What is the solution to this initial-value problem?

Source: Adapted from A. Flynn et al., "Effectiveness of Radiolabelled Antibodies for Radio-Immunotherapy in a Colorectal Xenograft Model: A Comparative Study Using the Linear-Quadratic Formulation," *International Journal of Radiation Biology* 77 (2001): 507–17.

- 7. Cancer progression** The development of many cancers, such as colorectal cancer, proceed through a series of precancerous stages. Suppose there are $n - 1$ precancerous stages before developing into cancer at stage n . A simple system of differential equations modeling this is

$$\begin{aligned} x_0' &= -u_0 x_0 \\ x_i' &= u_{i-1} x_{i-1} - u_i x_i \\ x_n' &= u_{n-1} x_{n-1} \end{aligned}$$

where x_i is the fraction of the population in state i , the u_i 's are positive constants, and $i = 1, \dots, n - 1$.

- Suppose $n = 2$. What is the system of differential equations for the three stages?
- Note that the variable x_2 does not appear in the equations for the rate of change of x_0 or x_1 . Consequently, we can solve the two-dimensional system for x_0 and x_1 separately. Do so, assuming that $x_0(0) = k$ and $x_1(0) = 0$.
- Use your solution for $x_1(t)$ obtained in part (b) to write a differential equation for $x_2(t)$.
- Solve the differential equation from part (c), assuming $x_2(0) = 0$.

- 8. Metapopulations** Consider a simple metapopulation in which subpopulation A grows at a per capita rate of $r_A = 1$

and subpopulation B declines at a per capita rate of $r_B = -1$. Suppose the per capita rate of movement between subpopulation patches is m in both directions. This gives

$$\begin{aligned}\frac{dx_A}{dt} &= (1 - m)x_A + mx_B \\ \frac{dx_B}{dt} &= -(1 + m)x_B + mx_A\end{aligned}$$

where x_A and x_B are the numbers of individuals in patches A and B, respectively.

- Classify the equilibrium at the origin.
- Find the general solution.
- What is the solution to the initial-value problem if $x_A(0) = 1$ and $x_B(0) = 0$?

9. Suppose a glass of cold water is sitting in a warm room and you place a coin at room temperature R into the glass. The coin gradually cools down while, at the same time, the glass of water warms up. Newton's law of cooling suggests the following system of differential equations to describe the process

$$\frac{dw}{dt} = -k_w(w - R) \quad \frac{dp}{dt} = -k_p(p - w)$$

where w and p are the temperatures of the water and coin (in $^{\circ}\text{C}$), respectively, and the k 's are positive constants.

- Explain the form of the system of differential equations and the assumptions that underlie them.
- Use a change of variables to obtain a homogeneous system.
- What is the general solution to the system you found in part (b)?
- What is the solution to the original initial-value problem if $w(0) = w_0$ and $p(0) = p_0$?

10. **Vaccine coverage** The project on page 479 explores an epidemiological model of vaccine coverage. For certain values of the constants the system of differential equations is

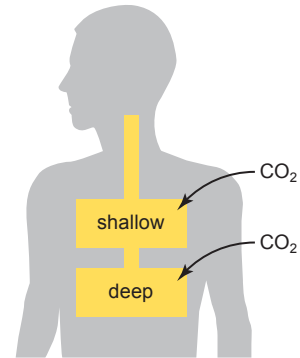
$$\begin{aligned}\frac{dN}{dt} &= 2(1 - p)N + 2(1 - p)V - N \\ \frac{dV}{dt} &= 2pV + 2pN - 3V\end{aligned}$$

where N and V are the numbers of nonvaccinated and vaccinated infected people, respectively, and p is the fraction of the population that is vaccinated.

- Classify the equilibrium at the origin for all values of the vaccination coverage p .
- What is the critical vaccination coverage above which the infection dies out?
- What is the general solution?
- What is the solution to the initial-value problem where $N(0) = k$ and $V(0) = 0$?

11. **Pulmonary air embolism** is a type of blood clot, which can occur during surgery, whereby part of the lung is no longer able to exchange CO_2 for O_2 . It is detected by a marked drop in the CO_2 concentration in exhaled air. A simple

model of this process involves two lung compartments: a deep and a shallow compartment, each of which removes CO_2 from the blood.



Using a for the rate of absorption of gas into each compartment from the blood, c as the concentration of CO_2 in this gas, and V as the volume of each compartment, we can model gas exchange during normal lung functioning as

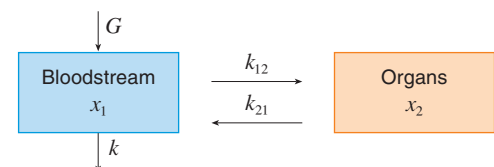
$$\frac{dx_1}{dt} = ac + a \frac{x_2}{V} - 2a \frac{x_1}{V} \quad \frac{dx_2}{dt} = ac - a \frac{x_2}{V}$$

where x_1 and x_2 are the amounts of CO_2 in the shallow and deep compartments, respectively. The concentration of CO_2 in the exhaled air at any time is given by x_1/V (see also Exercise 15 in the Review Section of this chapter).

- What is the equilibrium amount of CO_2 in the two compartments during normal lung functioning?
- If an embolism occurs in the deep lung, the equation for x_2 becomes $dx_2/dt = -ax_2/V$ because CO_2 is no longer entering this compartment. Use a change of variables to obtain a homogeneous system of differential equations for gas exchange during an embolism.
- The equilibrium values of x_1 and x_2 from part (a) can be used as the initial condition for the system of differential equations in part (b) to obtain an initial-value problem for the gas exchange once an embolism occurs. What is its solution?
- Use the solution found in part (c) to obtain a solution in term of the original variables x_1 and x_2 .

Source: Adapted from S. Cruickshank, *Mathematics and Statistics in Anaesthesia*. (New York: Oxford University Press, USA, 2004).

12. **Systemic lupus erythematosus** is an autoimmune disease in which some immune molecules, called antibodies, target DNA instead of pathogens. This can be treated by injecting drugs that absorb the offending antibodies. The antibodies are found in both the bloodstream and in organs, and this can be modeled using a two-compartment model:



A system of differential equations describing the amount of antibody in each compartment is

$$\begin{aligned}\frac{dx_1}{dt} &= G + k_{21}x_2 - k_{12}x_1 - kx_1 \\ \frac{dx_2}{dt} &= k_{12}x_1 - k_{21}x_2\end{aligned}$$

where G is the rate of generation of antibodies, k is the rate at which the drug treatment removes antibody from the bloodstream, and k_{ij} is the rate of flow of antibody from compartment i to j . The variables x_1 and x_2 are the amounts of antibody in the bloodstream and organs, respectively,

measured in μg . (See also Exercise 16 in the Review Section of this chapter.)

- Use a change of variables to obtain a homogeneous system of differential equations describing the situation.
- What is the general solution to the differential equations in part (a)?
- What is the general solution obtained in part (b) in terms of the original variables x_1 and x_2 ?

Source: Adapted from K. Suzuki et al., "Anti-DNA Antibody Kinetics Following Selective Removal by Adsorption using Dextran Sulphate Cellulose Columns in Patients with Systemic Lupus Erythematosus," *Journal of Clinical Apheresis* 11 (1996): 16–22.

PROJECT Pharmacokinetics of Antimicrobial Dosing

The term **pharmacokinetics** refers to the change in drug concentration within the body during treatment. Figure 1 gives an example of the serum concentration (in $\mu\text{g}/\text{mL}$) of the antimicrobial drug panobacumab in a patient during three consecutive infusions. The concentration increases nearly instantaneously during an infusion and it then declines through metabolism until the next infusion occurs.

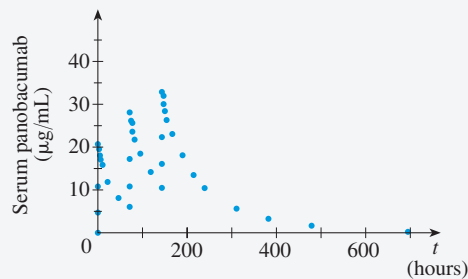


FIGURE 1

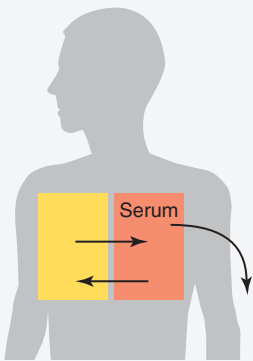


FIGURE 2

Mathematical models are routinely used to determine the drug dose and time between infusions required to achieve a desired peak serum level. A common model is a two-compartment mixing model. One compartment is the blood serum, and the other is the remainder of the body. The drug is infused directly into the serum and is metabolized from there. It also flows back and forth between the serum and the rest of the body (see Figure 2). Following the approach used in Exercises 7.4.45–48, we can obtain the following system of linear differential equations:

$$\begin{aligned}(1) \quad \frac{dx_1}{dt} &= \frac{k_{21}}{V_2}x_2 - \frac{k_{12}}{V_1}x_1 - \frac{k_{10}}{V_1}x_1 \\ \frac{dx_2}{dt} &= \frac{k_{12}}{V_1}x_1 - \frac{k_{21}}{V_2}x_2\end{aligned}$$

where x_1 and x_2 are the amounts in the serum and body, respectively (in $\mu\text{g}/\text{mL}$), k_{ij} are rate constants, and V_i is the effective volume of compartment i (in mL). The initial condition for the model is $x_1(0) = d$ and $x_2(0) = 0$, where d is the drug dose that is infused. Models like (1) often provide an excellent fit to data, as shown in Figure 3.

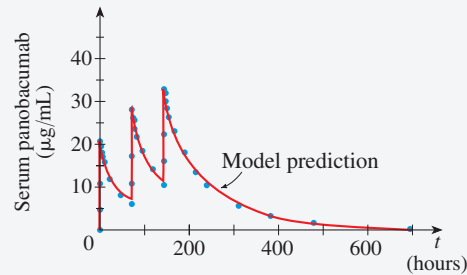


FIGURE 3

1. Explain the terms in Equations 1 and the assumptions that underlie them.
2. As an example, suppose that $k_{10}/V_1 = 2$, $k_{12}/V_1 = 1$, and $k_{21}/V_2 = 2$. Find the general solution to Equations 1.
3. Show that the general solution we obtained in Problem 2 can be written as $\varphi(t) \cdot \mathbf{c}$, where

$$\varphi(t) = \begin{bmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} \quad \mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

with λ_i and $\mathbf{v}_i$ being the eigenvalues and eigenvectors of the coefficient matrix.

4. Using our result from Problem 3, show that the solution to the initial-value problem can be written as $\mathbf{x}(t) = B(t)\mathbf{x}_0$, where $B(t) = \varphi(t)\varphi^{-1}(0)$ and $\mathbf{x}_0$ is the vector of initial values (that is, its components are d and 0).
5. If drug infusions continue repeatedly at fixed intervals, the level of drug in the serum eventually displays periodic behavior, as shown in Figure 4. Suppose τ is the time between infusions. If the amount of drug in the serum and body immediately after an infusion is denoted by the vector $\mathbf{y}$, then from Problem 4 the amounts immediately before the next infusion are $B(\tau)\mathbf{y}$. Immediately after the next infusion the amounts are $B(\tau)\mathbf{y} + \mathbf{x}_0$. (Why?) Further, if the amounts in each compartment are displaying periodic behavior, then immediately after infusion they must be at the same level as they were immediately after the previous infusion; that is, $B(\tau)\mathbf{y} + \mathbf{x}_0 = \mathbf{y}$. Use this last equation to find the function relating peak serum level to the dose and infusion constants d and τ .

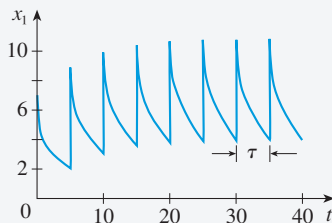


FIGURE 4

10.4 Systems of Nonlinear Differential Equations

Systems of linear differential equations form the basis for analyzing the stability properties of equilibria of systems of nonlinear differential equations. The approach is illustrated in this section. As with the rest of this chapter, we focus on two-dimensional systems.

Linear and Nonlinear Differential Equations

In the preceding sections we studied autonomous systems of differential equations having the form

$$(1) \quad \frac{dx_1}{dt} = a_{11}x_1 + a_{12}x_2 + g_1 \quad \frac{dx_2}{dt} = a_{21}x_1 + a_{22}x_2 + g_2$$

In the development of the general techniques we saw that it was sufficient to analyze systems in which $g_1 = 0$ and $g_2 = 0$.

The equations in (1) are referred to as a system of linear differential equations because the right side of each differential equation is a linear function of the dependent variables. Many of the systems introduced in Sections 7.5 and 7.6 do not have this form. For example, the predator-prey equations on page 469 involve the product of the two dependent variables. Similarly, the Fitzhugh-Nagumo equations on page 475 involve the third power of one of the dependent variables.

Systems of autonomous differential equations that are not linear functions of the dependent variables are referred to as **systems of autonomous nonlinear differential equations**. They have the more general form

$$(2) \quad \begin{aligned} \frac{dx_1}{dt} &= f_1(x_1, x_2) \\ \frac{dx_2}{dt} &= f_2(x_1, x_2) \end{aligned}$$

where f_1 and f_2 are arbitrary functions of the dependent variables. We will assume that both f_1 and f_2 have continuous first partial derivatives.

The nullclines of linear systems of differential equations are straight lines and therefore such systems typically have a single equilibrium. As we saw in Chapter 7, nonlinear systems often have several nonlinear nullclines and therefore they often have several equilibria as well. In general, it is not possible to solve systems of equations like (2) and so we focus on finding equilibria and determining their stability properties.

(3) Definition An **equilibrium** of Equations 2 is a pair of values $(\hat{x}_1, \hat{x}_2)$ such that both $dx_1/dt = 0$ and $dx_2/dt = 0$ when $x_1 = \hat{x}_1$ and $x_2 = \hat{x}_2$. The functions $x_1(t)$ and $x_2(t)$ are sometimes written in vector notation as $\mathbf{x}(t)$ and the equilibria as $\hat{\mathbf{x}}$.

Stability of equilibria in systems of nonlinear differential equations is defined in a way analogous to that for single nonlinear differential equations (see Section 7.2).

Definition An equilibrium $\hat{\mathbf{x}}$ of the system of differential equations (2) is **locally stable** if $\mathbf{x}$ approaches the value $\hat{\mathbf{x}}$ as $t \rightarrow \infty$ for all initial conditions sufficiently close to $\hat{\mathbf{x}}$.

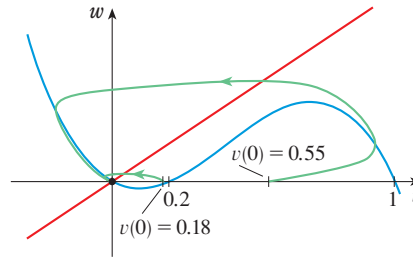
■ Local Stability Analyses

Recall the Fitzhugh-Nagumo equations for the electrical potential of a neuron from page 475. Using $v(t)$ for the neuron's electrical potential at time t and $w(t)$ to denote the magnitude of ion exchange, we had

$$(4) \quad \begin{aligned} \frac{dv}{dt} &= (v - a)(1 - v)v - w \\ \frac{dw}{dt} &= bv - cw \end{aligned}$$

Both v and w can be positive or negative. In the example on page 474 we used the values $a = 0.2$, $b = 0.01$, and $c = 0.04$. Figure 1 displays the nullclines, along with two solution curves in the phase plane.

FIGURE 1
The v -nullcline is shown in blue, w -nullcline in red, and solution curves in green.



Notice that the w -nullcline is linear, with a slope of $b/c = \frac{1}{4}$, because the differential equation for w is linear. The v -nullcline is nonlinear because the differential equation for v is nonlinear—it is the graph of a cubic polynomial in v . The origin, $(v, w) = (0, 0)$, is the only equilibrium because it is the only place at which these nullclines intersect.

The two solution curves plotted on Figure 1 both ultimately move toward the origin but it is not possible to determine from the phase plane if this is also true for other initial conditions. Consequently, we need a more precise approach.

To begin, suppose the initial condition is very close to the origin. By zooming in on the origin in Figure 1 and removing the solution curves, we get Figure 2. Locally (that is, near the origin) the cubic nullcline now looks approximately linear. In fact, as we know from Section 3.8, we can approximate this cubic polynomial near the origin with a line. If we define $g(v) = (v - a)(1 - v)v$, then the cubic nullcline is defined by the equation $w = g(v)$. Now, near the origin we can approximate $g(v)$ as

$$(5) \quad g(v) \approx g(0) + g'(0)v = -av$$

Therefore the nullcline is approximately given by $w \approx -av$, as shown in Figure 3.

We can also use approximation (5) in Equations 4 to obtain the following system of linear differential equations:

$$(6) \quad \begin{aligned} \frac{dv}{dt} &= -av - w \\ \frac{dw}{dt} &= bv - cw \end{aligned}$$

The lines drawn in Figure 3 are precisely the nullclines of Equations 6. Equations 6 are referred to as the linear approximation to Equations 4 near the equilibrium $(v, w) = (0, 0)$ or, equivalently, the *linearization* of Equations 4 near this point.

Using the values $a = 0.2$, $b = 0.01$, and $c = 0.04$, we can write system (6) in matrix notation as

$$(7) \quad \frac{d\mathbf{x}}{dt} = A\mathbf{x} \quad \text{with} \quad A = \begin{bmatrix} -0.2 & -1 \\ 0.01 & -0.04 \end{bmatrix}$$

The eigenvalues of A are $\lambda = -0.12 \pm 0.06i$, indicating that the origin in system (7) is a stable spiral. But what does this mean in terms of the local stability of the origin for system (4)? Provided that the initial condition of system (4) is close enough to the origin, the linearization given by system (7) should be a reasonable approximation. Moreover, the origin in this linear system is a stable spiral. Therefore we might expect all solutions

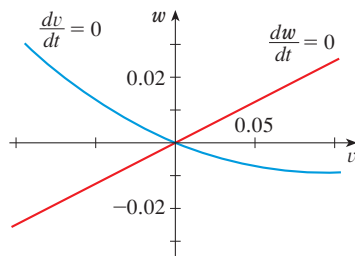


FIGURE 2

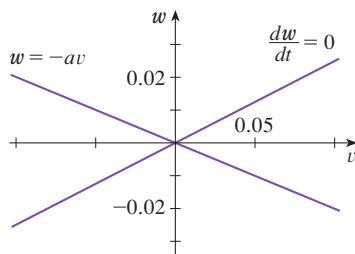


FIGURE 3

of system (4) to approach the origin in a spiraling fashion as well, provided they start sufficiently close to the origin (see Figure 4).

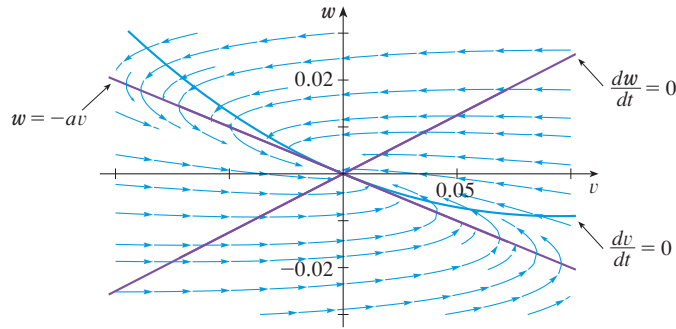


FIGURE 4

Solution curves for the linear system in Equations 7, along with its nullclines in purple. The v -nullcline of the original nonlinear model is blue.

Linearization

The preceding calculations can be extended to the following general system of autonomous nonlinear differential equations:

$$(8) \quad \begin{aligned} \frac{dx_1}{dt} &= f_1(x_1, x_2) \\ \frac{dx_2}{dt} &= f_2(x_1, x_2) \end{aligned}$$

Let's consider an arbitrary equilibrium of Equations 8, denoted $(\hat{x}_1, \hat{x}_2)$. Our derivation will follow that of Section 7.2 for one-variable, nonlinear differential equations.

The equilibrium of interest need no longer be the origin and therefore we first change variables so that we can again focus attention on the origin. Defining $\varepsilon_1 = x_1 - \hat{x}_1$ and $\varepsilon_2 = x_2 - \hat{x}_2$, and noting that

$$\frac{d\varepsilon_i}{dt} = \frac{d}{dt}[x_i - \hat{x}_i] = \frac{dx_i}{dt}$$

we have, from system (8),

$$(9) \quad \begin{aligned} \frac{d\varepsilon_1}{dt} &= f_1(\varepsilon_1 + \hat{x}_1, \varepsilon_2 + \hat{x}_2) \\ \frac{d\varepsilon_2}{dt} &= f_2(\varepsilon_1 + \hat{x}_1, \varepsilon_2 + \hat{x}_2) \end{aligned}$$

The variables ε_1 and ε_2 are the deviations of x_1 and x_2 from their equilibrium values, and therefore they have an equilibrium at $(\varepsilon_1, \varepsilon_2) = (0, 0)$. Furthermore, this equilibrium is locally stable if and only if the equilibrium $(\hat{x}_1, \hat{x}_2)$ of system (8) is locally stable.

The next step is to obtain the linear approximation of the right side of each differential equation in (9), near the origin $(\varepsilon_1, \varepsilon_2) = (0, 0)$. To do so, we need to use the two-variable tangent plane approximation in Equation 4 on page 599. We have

$$(10) \quad f_1(\varepsilon_1 + \hat{x}_1, \varepsilon_2 + \hat{x}_2) \approx f_1(\hat{x}_1, \hat{x}_2) + \frac{\partial f_1(\hat{x}_1, \hat{x}_2)}{\partial x_1} \varepsilon_1 + \frac{\partial f_1(\hat{x}_1, \hat{x}_2)}{\partial x_2} \varepsilon_2$$

and

$$(11) \quad f_2(\varepsilon_1 + \hat{x}_1, \varepsilon_2 + \hat{x}_2) \approx f_2(\hat{x}_1, \hat{x}_2) + \frac{\partial f_2(\hat{x}_1, \hat{x}_2)}{\partial x_1} \varepsilon_1 + \frac{\partial f_2(\hat{x}_1, \hat{x}_2)}{\partial x_2} \varepsilon_2$$

Equations 10 and 11 can be further simplified by noting that $f_1(\hat{x}_1, \hat{x}_2) = 0$ and $f_2(\hat{x}_1, \hat{x}_2) = 0$ because the point $(\hat{x}_1, \hat{x}_2)$ is an equilibrium of system (8). Substituting the resulting expressions into the right side of system (9) gives

$$\begin{aligned}\frac{d\varepsilon_1}{dt} &\approx \frac{\partial f_1(\hat{x}_1, \hat{x}_2)}{\partial x_1} \varepsilon_1 + \frac{\partial f_1(\hat{x}_1, \hat{x}_2)}{\partial x_2} \varepsilon_2 \\ \frac{d\varepsilon_2}{dt} &\approx \frac{\partial f_2(\hat{x}_1, \hat{x}_2)}{\partial x_1} \varepsilon_1 + \frac{\partial f_2(\hat{x}_1, \hat{x}_2)}{\partial x_2} \varepsilon_2\end{aligned}$$

or, in matrix notation,

$$(12) \quad \frac{d\mathbf{\varepsilon}}{dt} \approx \begin{bmatrix} \frac{\partial f_1(\hat{x}_1, \hat{x}_2)}{\partial x_1} & \frac{\partial f_1(\hat{x}_1, \hat{x}_2)}{\partial x_2} \\ \frac{\partial f_2(\hat{x}_1, \hat{x}_2)}{\partial x_1} & \frac{\partial f_2(\hat{x}_1, \hat{x}_2)}{\partial x_2} \end{bmatrix} \mathbf{\varepsilon}$$

where $\mathbf{\varepsilon}$ is the vector whose components are ε_1 and ε_2 . Equation 12 is the **linearization** of Equations 8 near the equilibrium point $(\hat{x}_1, \hat{x}_2)$.

Finally, the key observation is that the entries of the matrix of coefficients in Equation 12 are evaluated at the equilibrium $(\hat{x}_1, \hat{x}_2)$. As a result, they do not involve the variables. Consequently, Equation 12 is an autonomous, homogeneous system of linear differential equations for $\mathbf{\varepsilon}(t)$. Although Equation 12 is an approximation, we might expect the eigenvalues of its coefficient matrix to determine the local stability of the equilibrium $(\varepsilon_1, \varepsilon_2) = (0, 0)$, and therefore the local stability of $(\hat{x}_1, \hat{x}_2)$. This is borne out by the following definition and theorem.

(13) Definition Consider the autonomous system of differential equations

$$\frac{dx_1}{dt} = f_1(x_1, x_2) \quad \frac{dx_2}{dt} = f_2(x_1, x_2)$$

The matrix

$$J(x_1, x_2) = \begin{bmatrix} \frac{\partial f_1(x_1, x_2)}{\partial x_1} & \frac{\partial f_1(x_1, x_2)}{\partial x_2} \\ \frac{\partial f_2(x_1, x_2)}{\partial x_1} & \frac{\partial f_2(x_1, x_2)}{\partial x_2} \end{bmatrix}$$

is called the **Jacobian** matrix. Note that J is a function of x_1 and x_2 because its entries are functions of x_1 and x_2 .

We then have the following theorem:

(14) Theorem (Local Stability) Suppose $(\hat{x}_1, \hat{x}_2)$ is an equilibrium of the system of differential equations in Definition 13. Let r be the largest eigenvalue of $J(\hat{x}_1, \hat{x}_2)$, or the largest real part of the eigenvalues if they are complex. If $r < 0$, then the equilibrium is locally stable. If $r > 0$, then the equilibrium is unstable. If $r = 0$, then the analysis is inconclusive.

Examples

EXAMPLE 1 | Lotka-Volterra competition equations In Example 7.6.1 we studied a pair of equations that describe competition between two species. These equations were obtained by extending the logistic equation for population growth in a single species. The equations are

$$(15) \quad \frac{dN_1}{dt} = \left(1 - \frac{N_1 + 2N_2}{1000}\right)N_1 \quad \frac{dN_2}{dt} = \left(1 - \frac{N_2 + N_1}{600}\right)N_2$$

where N_1 and N_2 are the population sizes of species 1 and 2. We identified four equilibria:

- (a) $\hat{N}_1 = 0, \quad \hat{N}_2 = 0$
- (b) $\hat{N}_1 = 0, \quad \hat{N}_2 = 600$
- (c) $\hat{N}_1 = 1000, \quad \hat{N}_2 = 0$
- (d) $\hat{N}_1 = 200, \quad \hat{N}_2 = 400$

Determine the local stability properties of each equilibrium.

SOLUTION The first step is to calculate the Jacobian matrix. Defining

$$f_1(N_1, N_2) = \left(1 - \frac{N_1 + 2N_2}{1000}\right)N_1 \quad f_2(N_1, N_2) = \left(1 - \frac{N_2 + N_1}{600}\right)N_2$$

we obtain

$$J = \begin{bmatrix} \frac{\partial f_1(N_1, N_2)}{\partial N_1} & \frac{\partial f_1(N_1, N_2)}{\partial N_2} \\ \frac{\partial f_2(N_1, N_2)}{\partial N_1} & \frac{\partial f_2(N_1, N_2)}{\partial N_2} \end{bmatrix} = \begin{bmatrix} 1 - \frac{N_1 + N_2}{500} & -\frac{N_1}{500} \\ -\frac{N_2}{600} & 1 - \frac{N_1 + 2N_2}{600} \end{bmatrix}$$

- (a) Evaluating the Jacobian at $\hat{N}_1 = 0, \hat{N}_2 = 0$ gives

$$J = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The eigenvalues of J are $\lambda_1 = 1$ and $\lambda_2 = 1$. Thus $r = 1$ in Theorem 14 and the equilibrium is unstable.

- (b) Evaluating the Jacobian at $\hat{N}_1 = 0, \hat{N}_2 = 600$ gives

$$J = \begin{bmatrix} -\frac{1}{5} & 0 \\ -1 & -1 \end{bmatrix}$$

The eigenvalues of J are $\lambda_1 = -\frac{1}{5}$ and $\lambda_2 = -1$. Thus $r = -\frac{1}{5}$ in Theorem 14 and the equilibrium is locally stable.

- (c) Evaluating the Jacobian at $\hat{N}_1 = 1000, \hat{N}_2 = 0$ gives

$$J = \begin{bmatrix} -1 & -2 \\ 0 & -\frac{2}{3} \end{bmatrix}$$

The eigenvalues of J are $\lambda_1 = -1$ and $\lambda_2 = -\frac{2}{3}$. Thus $r = -\frac{2}{3}$ in Theorem 14 and the equilibrium is locally stable.

(d) Evaluating the Jacobian at $\hat{N}_1 = 200$, $\hat{N}_2 = 400$ gives

$$J = \begin{bmatrix} -\frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{3} & -\frac{2}{3} \end{bmatrix}$$

The eigenvalues of J are $\lambda_1 = -1$ and $\lambda_2 = \frac{2}{15}$. Thus $r = \frac{2}{15}$ in Theorem 14 and the equilibrium is unstable.

The preceding analysis matches the phase plane analysis in Figure 7.6.5 by showing that the only stable equilibria are those where a single species is present. This is referred to as *competitive exclusion*. In this example, the species that ultimately survives is determined by the initial size of each population. ■

The following theorem is often useful when evaluating local stability. It follows directly from a combination of Theorem 10.2.15 and Theorem 10.2.16.

(16) Theorem Suppose $(\hat{x}_1, \hat{x}_2)$ is an equilibrium of the system of differential equations in Definition 13. If

- (i) $\det J(\hat{x}_1, \hat{x}_2) > 0$, and
- (ii) $\text{trace } J(\hat{x}_1, \hat{x}_2) < 0$

then the equilibrium is locally stable. If the inequalities in either (i), (ii), or both are reversed, then the equilibrium is unstable. (Recall that the trace of a matrix is the sum of its diagonal entries.)

EXAMPLE 2 | Kermack-McKendrick infectious disease model In Exercise 7.6.24 we studied a pair of equations that describe the spread of an infectious disease in the human population. The equations are

$$(17) \quad \frac{dS}{dt} = \theta - \gamma S - \beta SI \quad \frac{dI}{dt} = \beta SI - \mu I$$

where S and I are the numbers of susceptible and infected people, respectively. The constant θ is the rate of arrival of susceptible people, γ is their per capita mortality rate, β is the transmission rate of the disease from infected to susceptible people, and μ is the per capita mortality rate of infected people. All constants are positive.

Definition 3 can be used to verify that the following are both equilibria:

- (a) $\hat{S} = \theta/\gamma$, $\hat{I} = 0$
- (b) $\hat{S} = \mu/\beta$, $\hat{I} = (\beta\theta - \gamma\mu)/\beta\mu$

Note that the second equilibrium is of biological interest only if $\beta\theta - \gamma\mu > 0$. Assuming $\beta\theta - \gamma\mu > 0$, determine the local stability properties of both equilibria.

SOLUTION The first step is to calculate the Jacobian matrix. Defining

$$f_1(S, I) = \theta - \gamma S - \beta SI \quad f_2(S, I) = \beta SI - \mu I$$

we obtain

$$J = \begin{bmatrix} \frac{\partial f_1(S, I)}{\partial S} & \frac{\partial f_1(S, I)}{\partial I} \\ \frac{\partial f_2(S, I)}{\partial S} & \frac{\partial f_2(S, I)}{\partial I} \end{bmatrix} = \begin{bmatrix} -\gamma - \beta I & -\beta S \\ \beta I & \beta S - \mu \end{bmatrix}$$

(a) Evaluating the Jacobian at $\hat{S} = \theta/\gamma$ and $\hat{I} = 0$ gives

$$J = \begin{bmatrix} -\gamma & -\frac{\beta\theta}{\gamma} \\ 0 & \frac{\beta\theta}{\gamma} - \mu \end{bmatrix}$$

The eigenvalues are therefore $\lambda_1 = -\gamma$ and $\lambda_2 = \beta\theta/\gamma - \mu$. The assumption that $\beta\theta - \gamma\mu > 0$ means that $\lambda_2 > 0$ and therefore this equilibrium is unstable. This illustrates that the disease will spread in the population.

(b) Evaluating the Jacobian at $\hat{S} = \mu/\beta$ and $\hat{I} = (\beta\theta - \gamma\mu)/\beta\mu$ gives

$$J = \begin{bmatrix} -\gamma - \frac{\beta\theta - \gamma\mu}{\mu} & -\mu \\ \frac{\beta\theta - \gamma\mu}{\mu} & 0 \end{bmatrix}$$

The eigenvalues of J can be readily calculated, but their signs are difficult to determine. Instead we make use of Theorem 16. We have

$$\det J = \beta\theta - \gamma\mu$$

and

$$\text{trace } J = -\gamma - \frac{\beta\theta - \gamma\mu}{\mu}$$

The assumption that $\beta\theta - \gamma\mu > 0$ means that $\det J > 0$ and $\text{trace } J < 0$. Therefore, from Theorem 16, this equilibrium is locally stable. At this stable equilibrium, part of the population will be infected and the remainder will be susceptible to infection. ■

EXAMPLE 3 | Lotka-Volterra predator-prey equations In Section 7.5 we studied a pair of equations that describe a predator and its prey. The equations are

$$(18) \quad \frac{dR}{dt} = rR - aRW \quad \frac{dW}{dt} = -kW + bRW$$

where R and W are the population sizes of prey and predator, respectively. The constant r is the per capita reproductive rate of prey, k is the per capita death rate of the predator, a is the rate of consumption of prey by predators, and b is the rate at which this consumption is converted into predator offspring. We identified two equilibria of these equations in Section 7.6, one of which was $(\hat{R}, \hat{W}) = (k/b, r/a)$. Evaluate the stability of this equilibrium.

SOLUTION Defining

$$f_1(R, W) = rR - aRW \quad f_2(R, W) = -kW + bRW$$

the Jacobian is

$$J = \begin{bmatrix} \frac{\partial f_1(R, W)}{\partial R} & \frac{\partial f_1(R, W)}{\partial W} \\ \frac{\partial f_2(R, W)}{\partial R} & \frac{\partial f_2(R, W)}{\partial W} \end{bmatrix} = \begin{bmatrix} r - aW & -aR \\ bW & -k + bR \end{bmatrix}$$

Evaluating this at the equilibrium gives

$$J = \begin{bmatrix} 0 & -\frac{ak}{b} \\ \frac{rb}{a} & 0 \end{bmatrix}$$

The eigenvalues of J are the complex conjugates $\lambda_1 = i\sqrt{rk}$ and $\lambda_2 = -i\sqrt{rk}$. The real part of these eigenvalues is 0 and therefore $r = 0$ in Theorem 14. As a result, the local stability analysis is inconclusive. Figure 7.5.4 on page 462 suggests that, in this case, the equilibrium is *neutrally stable*, that is, solution curves move neither toward the equilibrium nor away from it as $t \rightarrow \infty$. This can, in fact, be proven, and you should consider how you might do this. Thus the predator and prey populations are predicted to undergo neverending oscillations. ■

EXERCISES 10.4

1–6 Each of the nonlinear systems has an equilibrium at $(\hat{x}_1, \hat{x}_2) = (0, 0)$. Find the linearization near this point.

1. $\frac{dx_1}{dt} = 4x_1 - 2x_1x_2$

$$\frac{dx_2}{dt} = -2x_2 + 8x_1x_2$$

2. $\frac{dx_1}{dt} = 4x_1(1 - 5x_1) - 2x_1x_2$

$$\frac{dx_2}{dt} = -2x_2 + 8x_1x_2$$

3. $\frac{dx_1}{dt} = \sin x_1 + x_1x_2 + 3x_2^2$

$$\frac{dx_2}{dt} = \cos x_2 - 1 + x_1(x_1 - 1) + 7x_2$$

4. $\frac{dx_1}{dt} = 5(1 + \cos x_1) + ax_1 - bx_2 - 10$

$$\frac{dx_2}{dt} = 3x_2 + bx_1x_2$$

5. $\frac{dx_1}{dt} = 1 + x_1^3 - \frac{1 + x_1}{1 + x_2}$

$$\frac{dx_2}{dt} = 2x_2 + x_1^2$$

6. $\frac{dx_1}{dt} = x_2 - \frac{2 + ax_1}{2 + bx_2} + \cos x_2$

$$\frac{dx_2}{dt} = \frac{2x_1}{1 + x_2} - ax_1$$

7–12 Find all equilibria. Then find the linearization near each equilibrium.

7. $\frac{dx_1}{dt} = -5x_1 + x_1x_2$

$$\frac{dx_2}{dt} = x_2 - 5x_1x_2$$

8. $\frac{dx_1}{dt} = x_2 - 5x_1x_2$

$$\frac{dx_2}{dt} = 2x_1 - 6x_1x_2$$

9. $\frac{dx_1}{dt} = x_1 - 6x_2^2 + x_1x_2$

$$\frac{dx_2}{dt} = 8x_1 + 4x_1x_2$$

10. $\frac{dx_1}{dt} = x_1 - 2x_1^2 - 6x_1x_2$

$$\frac{dx_2}{dt} = 2x_2 - 8x_2^2 - 2x_1x_2$$

11. $\frac{dx_1}{dt} = e^{-x_1}(x_1 - x_2)$

$$\frac{dx_2}{dt} = x_1 - x_2^2 + 2x_1x_2$$

12. $\frac{dx_1}{dt} = \ln x_1 - x_2$

$$\frac{dx_2}{dt} = x_1(1 - x_1 - x_2)$$

13–18 A Jacobian matrix and two equilibria are given. Determine if each is locally stable, unstable, or if the analysis is inconclusive.

$$13. J = \begin{bmatrix} (x_1 - 2)x_2 + x_1x_2 & x_1(x_1 - 2) \\ 0 & -1 + 2x_2 \end{bmatrix}$$

- (i) $\hat{x}_1 = 0, \hat{x}_2 = 2$
 (ii) $\hat{x}_1 = 2, \hat{x}_2 = -1$

$$14. J = \begin{bmatrix} -1 + 2x_1 & 0 \\ 0 & -\frac{1}{3} + 2x_2 \end{bmatrix}$$

- (i) $\hat{x}_1 = -1, \hat{x}_2 = 0$
 (ii) $\hat{x}_1 = 2, \hat{x}_2 = \frac{1}{3}$

$$15. J = \begin{bmatrix} 1 - \cos x_2 & (x_1 - 1)\sin x_2 \\ \cos x_1 & -\sin 1 \end{bmatrix}$$

- (i) $\hat{x}_1 = 0, \hat{x}_2 = 0$
 (ii) $\hat{x}_1 = 1, \hat{x}_2 = 1$

$$16. J = \begin{bmatrix} 2x_1 & -\sin x_2 \\ \cos x_1 & 0 \end{bmatrix}$$

- (i) $\hat{x}_1 = 1, \hat{x}_2 = -\pi$
 (ii) $\hat{x}_1 = 1, \hat{x}_2 = \pi$

$$17. J = \begin{bmatrix} -\frac{1}{2 + x_2} & -1 + \frac{x_1}{(2 + x_2)^2} \\ \frac{x_2}{(1 + x_1)^2} & -1 - \frac{1}{1 + x_1} \end{bmatrix}$$

- (i) $\hat{x}_1 = 0, \hat{x}_2 = 0$
 (ii) $\hat{x}_1 = -2, \hat{x}_2 = -1 - \sqrt{3}$

$$18. J = \begin{bmatrix} -\frac{1}{1 + x_2} & \frac{x_1}{(1 + x_2)^2} \\ -1 + \frac{x_2}{(1 + x_1)^2} & -\frac{1}{1 + x_1} \end{bmatrix}$$

- (i) $\hat{x}_1 = -2, \hat{x}_2 = -2$
 (ii) $\hat{x}_1 = \frac{1}{2}, \hat{x}_2 = -\frac{3}{4}$

19–23 Find all equilibria and determine their local stability properties.

19. $x' = x(3 - x - y), \quad y' = y(2 - x - y)$
 20. $p' = p(1 - p - q), \quad q' = q(2 - 3p - q)$
 21. $n' = n(1 - 2m), \quad m' = m(2 - 2n - m)$
 22. $x' = x(2 - x), \quad y' = y(3 - y)$

$$23. p' = -p^2 + q - 1, \quad q' = q(2 - p - q)$$

24–25 Find all equilibria and determine their stability properties. Your answer might be a function of the constant a .

24. $x' = -xy + y + ax, \quad y' = 2y - xy$
 25. $x' = ax^2 + ay - x, \quad y' = x - y, \quad a \neq 0$

26. Cell cycle In Exercise 7.Review.25 a model for the cell cycle was introduced. It modeled the concentrations of a molecule called MPF (maturation promoting factor) and another molecule called cyclin. MPF production is stimulated by cyclin, but the presence of MPF also inhibits its own production. Using M and C to denote the concentrations of these two biomolecules (in mg/mL), the model for their interaction is

$$\begin{aligned} \frac{dM}{dt} &= \alpha C + \beta CM^2 - \frac{\gamma M}{1 + M} \\ \frac{dC}{dt} &= \delta - M \end{aligned}$$

- (a) Suppose that $\alpha = 2, \beta = 1, \gamma = 10$, and $\delta = 1$. Find the only equilibrium.
 (b) Calculate the Jacobian matrix.
 (c) Determine the local stability properties of the equilibrium found in part (a) using the Jacobian from part (b).
 (d) Describe how M and C change near the equilibrium point.

Source: Adapted from R. Norel et al., "A Model for the Adjustment of the Mitotic Clock by Cyclin and MPF Levels," *Science* 251 (1991): 1076–78.

27. Competition-colonization models In Exercise 7.Review.23 a metapopulation model for two species was introduced. The equations were

$$\begin{aligned} \frac{dp_1}{dt} &= c_1 p_1 (1 - p_1) - m_1 p_1 \\ \frac{dp_2}{dt} &= c_2 p_2 (1 - p_1 - p_2) - m_2 p_2 - c_1 p_1 p_2 \end{aligned}$$

where p_i is the fraction of patches occupied by species i , and c_i and m_i are the species-specific rates of colonization and extinction of patches, respectively. These equations assume that any patch has at most one species, and species 2 patches can be taken over by species 1, but not vice versa.

- (a) Suppose that $m_1 = m_2 = 3, c_1 = 5$, and $c_2 = 30$. Find all equilibria.
 (b) Calculate the Jacobian matrix.
 (c) Determine the local stability properties of each equilibrium found in part (a) using the Jacobian from part (b).
 (d) Are the species predicted to be able to coexist at a stable equilibrium?

- 28. Gene regulation** The model of gene regulation from Section 10.3 is often extended to nonlinear gene regulation by specifying a nonlinear function for how the concentration of protein in a cell affects mRNA production. One such example, called an **auto-repression** model, is

$$\frac{dm}{dt} = \frac{1}{1+p} - m \quad \frac{dp}{dt} = m - p$$

- Find all equilibria.
- Calculate the Jacobian matrix.
- Only one equilibrium found in part (a) is of biological interest. Determine its local stability properties using the Jacobian from part (b).
- Describe how m and p change near the equilibrium point.

29–31 Consumer resource models often have the following general form

$$R' = f(R) - g(R, C) \quad C' = \varepsilon g(R, C) - h(C)$$

where R is the number of individuals of the resource and C is the number of consumers. The function $f(R)$ gives the rate of replenishment of the resource, $g(R, C)$ describes the rate of consumption of the resource, and $h(C)$ is the rate of loss of the consumer. The constant ε , where $0 < \varepsilon < 1$, is the conversion efficiency of resources into consumers. Find all equilibria of the following examples and determine their stability properties.

- A chemostat** is an experimental consumer-resource system. If the resource is not self-reproducing, then it can be modeled by choosing $f(R) = \theta$, $g(R, C) = bRC$, and $h(C) = \mu C$. Suppose $\theta = 2$, $b = 1$, $\varepsilon = 1$, and $\mu = 1$.
- A model for self-reproducing resources is obtained by choosing $f(R) = rR$, $g(R, C) = bRC$, and $h(C) = \mu C$. Suppose $r = 2$, $b = 1$, $\varepsilon = 1$, and $\mu = 1$.
- A model for self-reproducing resources with limited growth is obtained by choosing $f(R) = rR(1 - R/K)$, $g(R, C) = bRC$, and $h(C) = \mu C$. Suppose $r = 2$, $K = 5$, $b = 1$, $\varepsilon = 1$, and $\mu = 1$.

- 32. The Kermack-McKendrick equations** describe the outbreak of an infectious disease. Using S and I to denote the number of susceptible and infected people in a population, respectively, the equations are

$$S' = -\beta SI \quad I' = \beta SI - \mu I$$

where β and μ are positive constants representing the transmission rate and rate of recovery.

- Verify that $\hat{I} = 0$, along with any value of S , is an equilibrium.
- Calculate the Jacobian matrix.
- Using your answer to part (b), determine how large S must be to guarantee that the disease will spread when rare.

- 33. The Michaelis-Menten equations** describe a biochemical reaction in which an enzyme E and substrate S bind to form a complex C . This complex can then either dissociate back into its original components or undergo a reaction in which a product P is produced along with the free enzyme: $E + S \rightleftharpoons C \rightarrow E + P$. This can be expressed by the differential equations

$$\frac{dx}{dt} = -k_f xyM + k_r(1 - y)M$$

$$\frac{dy}{dt} = -k_f xyM + k_r(1 - y)M + k_{cat}(1 - y)M$$

$$\frac{dz}{dt} = k_{cat}(1 - y)M$$

where M is the total number of enzymes (both free and bound), x and z are the numbers of substrate and product molecules, y is the fraction of the enzyme pool that is free, and the k_i 's are positive constants.

- Although this is a system of three differential equations, x and y can be analyzed separately. Explain why.
- Find the only equilibrium.
- Calculate the Jacobian matrix.
- Determine the local stability properties of the equilibrium.

- 34. Stability of Caribbean reefs** Coral and macroalgae compete for space when colonizing Caribbean reefs. A modification of the model in Exercise 27 has been used to describe this process. The equations are

$$\frac{dM}{dt} = \gamma M(1 - M) - \frac{gM}{1 - C}$$

$$\frac{dC}{dt} = rC(1 - M - C) - \gamma CM - dC$$

where M is the fraction of the reef occupied by macroalgae, C is the fraction occupied by coral, r is the colonization rate of empty space by coral, d is the death rate of coral, γ is the rate of colonization by macroalgae (in both empty space and space occupied by coral), and g is a constant governing the death rate of macroalgae. Notice that the per capita death rate of macroalgae decreases as coral cover increases.

- Suppose that $r = 3$, $d = 1$, $\gamma = 2$, and $g = 1$. Find all equilibria. There are five, but only four of them are biologically relevant.
- Calculate the Jacobian matrix.
- Determine the local stability properties of the four relevant equilibria found in part (a).
- In part (c) you should find two equilibria that are locally stable. What do they represent in terms of the structure of the reef?

Source: Adapted from P. Mumby et al., "Thresholds and the Resilience of Caribbean Coral Reefs," *Nature* 450 (2007): 98–101.

- 35. Fitzhugh-Nagumo equations** Consider the following alternative form of the Fitzhugh-Nagumo equations:

$$\frac{dv}{dt} = (v - a)(1 - v)v - w \quad \frac{dw}{dt} = \varepsilon(v - w)$$

where $\varepsilon > 0$ and $0 < a < 1$.

- Verify that the origin is an equilibrium.
- Calculate the Jacobian matrix.
- Determine the local stability properties of the origin as a function of the constants.

Chapter 10 REVIEW

CONCEPT CHECK

- What is the difference between an autonomous and a nonautonomous system of differential equations?
- What is an equilibrium of a system of differential equations?
- Explain the difference between local and global stability in systems of differential equations.
- What is the difference between the solution of an initial-value problem and the general solution of a system of differential equations?
- What does the Existence and Uniqueness Theorem tell us about homogeneous systems of linear, autonomous differential equations?
- Explain the Superposition Principle.
- Explain the difference between nullclines and eigenvectors in systems of linear autonomous differential equations.
- What is the linearization of a system of nonlinear differential equations?
- Explain what a Jacobian matrix is.
- What do the eigenvalues of a Jacobian matrix from a system of nonlinear differential equations tell us?

Answers to the Concept Check can be found on the back endpapers.

TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

1. The following system of differential equations is linear:

$$\begin{aligned} \frac{dx}{dt} &= 3x + y + 3 \\ \frac{dy}{dt} &= 2xy - x + 5 \end{aligned}$$

2. The following system of differential equations is homogeneous:

$$\begin{aligned} \frac{dx_1}{dt} &= a(t)x_1 + b(t)x_2 \\ \frac{dx_2}{dt} &= c(t)x_1 + d(t)x_2 \end{aligned}$$

3. The nullclines of a two-variable system of linear differential equations must be straight lines.

- If an equilibrium of an autonomous system of linear, homogeneous differential equations is locally stable, then it is globally stable as well.
- A saddle equilibrium of an autonomous system of linear, homogeneous differential equations is always unstable.
- A stable node must have two eigenvalues of the same sign.
- Spirals have complex eigenvalues.
- Stable equilibria of systems of linear autonomous differential equations must have eigenvalues with negative real parts.
- Nonlinear systems of differential equations have at most one equilibrium.
- If the Jacobian matrix for a nonlinear system of differential equations has eigenvalues with negative real parts at an equilibrium, then this equilibrium is globally stable.

EXERCISES

1–4 Is the system linear or nonlinear?

1. $p' = 2q - 1$, $q' = q^2 - q - p$
2. $x' = 5 - 2x - y$, $y' = 2x - y$
3. $z' = tz - 2w$, $w' = z - w - 1$
4. $x' = -(x - 2) \ln(xy)$, $y' = e^x(x - y)$

5–8 Show that $x_1(t)$ and $x_2(t)$ are solutions to the initial-value problem $d\mathbf{x}/dt = A\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$.

5. $A = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}$ $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
 $x_1(t) = \cos t - 3 \sin t$, $x_2(t) = 2 \cos t - \sin t$

6. $A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$ $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$
 $x_1(t) = 2e^{2t} - e^{-t}$, $x_2(t) = 4e^{2t} + e^{-t}$

7. $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ $\mathbf{x}_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
 $x_1(t) = 2e^t - te^t$, $x_2(t) = e^t$

8. $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $\mathbf{x}_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
 $x_1(t) = \frac{1}{2} + \frac{3}{2}e^{2t}$, $x_2(t) = -\frac{1}{2} + \frac{3}{2}e^{2t}$

9–12 Solve the initial-value problem $d\mathbf{x}/dt = A\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$.

9. $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

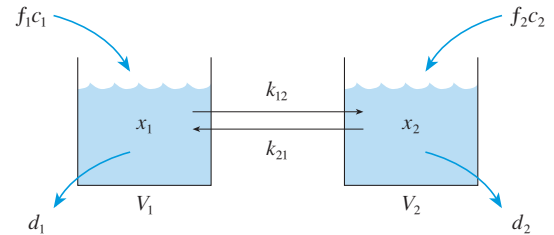
10. $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$ $\mathbf{x}_0 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

11. $A = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}$ $\mathbf{x}_0 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

12. $A = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$ $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

13–18 Two-compartment mixing problems are similar to the mixing problems in Exercises 7.4.45–48, except there are two connected tanks of fixed capacities filled with a well-mixed solution of some substance (for example, salt). Solution of a given concentration can enter each tank at a fixed rate, and solu-

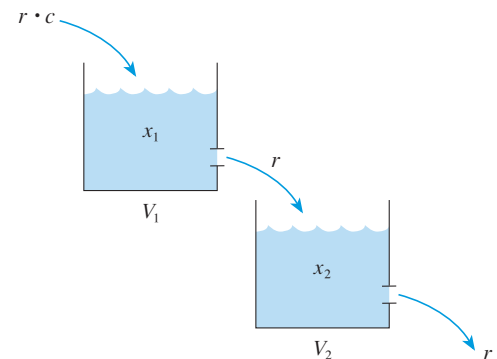
tion also flows back and forth between the tanks. Solution can also leave each tank at a fixed rate.



Using $x_1(t)$ and $x_2(t)$ to denote the amount of substance in each tank at time t , we obtain the following system of linear differential equations:

$$\begin{aligned} \frac{dx_1}{dt} &= f_1c_1 - k_{12} \frac{x_1}{V_1} + k_{21} \frac{x_2}{V_2} - d_1 \frac{x_1}{V_1} \\ \frac{dx_2}{dt} &= f_2c_2 - k_{21} \frac{x_2}{V_2} + k_{12} \frac{x_1}{V_1} - d_2 \frac{x_2}{V_2} \end{aligned}$$

13. Two tanks containing a salt solution are connected as in the figure:



Suppose that $x_1(t)$ and $x_2(t)$ denote the amount of salt (in grams) in each tank at time t (in minutes), and $r = 8$ L/min, $c = 2$ g/L, $V_1 = 50$ L, and $V_2 = 25$ L. The initial amounts of salt in each tank are $x_1(0) = 8$ g and $x_2(0) = 0$ g.

- (a) What is the initial-value problem describing this system?
- (b) Solve the initial-value problem from part (a).
- (c) What is the maximum amount of salt that is ever in tank 2?

14. **Hemodialysis** In Exercise 10.3.2 we modeled the process of hemodialysis, in which a machine filters urea from a patient's blood. The system of differential equations was

$$\begin{aligned} \frac{dc}{dt} &= -\frac{K}{V}c + ap - bc \\ \frac{dp}{dt} &= -ap + bc \end{aligned}$$

where c and p are the urea concentrations in the blood and in other parts of the body, respectively (in mg/mL). Explain how these equations can be viewed as a special case of a general two-compartment mixing problem.

- 15. Pulmonary air embolism** In Exercise 10.3.11 we modeled a pulmonary air embolism during surgery by viewing the lung as having two compartments: a deep and a shallow compartment, each of which removes CO_2 from the blood. The system of differential equations was

$$\begin{aligned}\frac{dx_1}{dt} &= ac + a \frac{x_2}{V} - 2a \frac{x_1}{V} \\ \frac{dx_2}{dt} &= ac - a \frac{x_2}{V}\end{aligned}$$

where x_1 and x_2 are the amounts of CO_2 in the shallow and deep compartments, respectively, a is the rate of absorption of gas into each compartment from the blood, c is the concentration of CO_2 in this gas, and V is the volume of each compartment. Explain how these equations can be viewed as a special case of a general two-compartment mixing problem.

- 16. Systemic lupus erythematosus** In Exercise 10.3.12 we modeled the amount of antibodies in the bloodstream and in other organs. The system of differential equations describing the amount of antibody in each compartment is

$$\begin{aligned}\frac{dx_1}{dt} &= G + k_{21}x_2 - k_{12}x_1 - kx_1 \\ \frac{dx_2}{dt} &= k_{12}x_1 - k_{21}x_2\end{aligned}$$

where x_1 and x_2 are the amounts in the blood and organs, respectively (in μg), G is the rate of generation of antibodies, k is the rate at which the drug treatment removes antibody from the bloodstream, and k_{ij} is the rate of flow of antibody from compartment i to j . Explain how these equations can be viewed as a special case of a general two-compartment mixing problem.

- 17.** Explain how the equations in the Project on page 458 can be viewed as a special case of a general two-compartment mixing problem.
- 18.** Explain how the equations in the Project on page 664 can be viewed as a special case of a general two-compartment mixing problem.

19–22 A Jacobian matrix and equilibrium are given. Is the equilibrium locally stable, unstable, or can you not tell?

19. $J = \begin{bmatrix} 0 & 2 \\ -1 & 2q - 1 \end{bmatrix}$ $\hat{p} = -\frac{1}{4}$, $\hat{q} = \frac{1}{2}$

20. $J = \begin{bmatrix} -2 - y & -x \\ y & x - 1 \end{bmatrix}$ $\hat{x} = \frac{5}{2}$, $\hat{y} = 0$

21. $J = \begin{bmatrix} 3z^2 - 8z + 3 & -2 \\ 1 & -1 \end{bmatrix}$ $\hat{z} = 1$, $\hat{w} = 0$

22. $J = \begin{bmatrix} \frac{2-x}{x} - \ln xy & \frac{2-x}{y} \\ e^x + e^x(x-y) & -e^x \end{bmatrix}$ $\hat{x} = 2$, $\hat{y} = 2$

23–26 Find all equilibria and determine their local stability properties.

23. $\frac{dx}{dt} = 5x - xy$, $\frac{dy}{dt} = 4y - y^2 - 2xy$

24. $\frac{dA}{dt} = 2A - AL$, $\frac{dL}{dt} = -5L + AL$

25. $\frac{dR}{dt} = 8R(1 - 2R) - RW$, $\frac{dW}{dt} = -2W + 2RW$

26. $\frac{dA}{dt} = 2A(1 - A) - AL$, $\frac{dL}{dt} = -5L + AL$

27. The **Lotka-Volterra competition equations** are

$$\frac{dN_1}{dt} = \left(1 - \frac{N_1 + \alpha N_2}{K_1}\right)N_1$$

$$\frac{dN_2}{dt} = \left(1 - \frac{N_2 + \beta N_1}{K_2}\right)N_2$$

where all constants α , β , K_1 , and K_2 are positive.

- What are the equilibria?
- Calculate the Jacobian matrix.
- Determine the local stability properties of all biologically feasible equilibria when $K_1 > \alpha K_2$ and $K_2 < \beta K_1$.
- Determine the local stability properties of all biologically feasible equilibria when $K_1 < \alpha K_2$ and $K_2 > \beta K_1$.
- Determine the local stability properties of all biologically feasible equilibria when $K_1 < \alpha K_2$ and $K_2 < \beta K_1$.
- Determine the local stability properties of all biologically feasible equilibria when $K_1 > \alpha K_2$ and $K_2 > \beta K_1$.

28. The **Rosenzweig-MacArthur model** is a consumer-resource model. A simplified version is

$$R' = R(K - R) - \frac{R}{a + R}C \quad C' = \frac{R}{a + R}C - bC$$

where R is the number of resource individuals and C is the number of consumers. Suppose that all constants are positive.

- What are the equilibria?
- Calculate the Jacobian matrix.
- Determine the values of the constants for which the extinction equilibrium is unstable.
- Determine the values of the constants for which the equilibrium with only the resource present is unstable.

- 29. Habitat destruction** The model of Exercise 10.4.27 can be extended to include the effects of habitat destruction. Suppose that only a fraction, h , of the patches are habitable ($0 < h < 1$). The equations become

$$\begin{aligned}\frac{dp_1}{dt} &= c_1 p_1 (h - p_1) - m_1 p_1 \\ \frac{dp_2}{dt} &= c_2 p_2 (h - p_1 - p_2) - m_2 p_2 - c_1 p_1 p_2\end{aligned}$$

- What are the equilibria?
- Calculate the Jacobian matrix.
- Determine the values of h for which the extinction equilibrium is stable.

Source: Adapted from S. Nee et al., “Dynamics of Metapopulations: Habitat Destruction and Competitive Coexistence,” *Journal of Animal Ecology* 61 (1992): 37–40.

- 30. Gene regulation** The model of gene regulation from Section 10.3 is often extended to nonlinear gene regulation by specifying a nonlinear function for how the concentration of protein in a cell affects mRNA production. One such example, called an **auto-activation** model, is

$$\begin{aligned}\frac{dm}{dt} &= \frac{2p}{1 + 2p} - m \\ \frac{dp}{dt} &= m - p\end{aligned}$$

- Find all equilibria.
- Calculate the Jacobian matrix.
- Determine the local stability properties of all equilibria.

- 31. Sterile insect technique** Sterile insects are sometimes released as a means of controlling insect populations. Fertile insects mate with the sterile individuals and therefore fail to produce offspring. The following differential equations model this idea:

$$\begin{aligned}\frac{df}{dt} &= af \frac{f}{f + s} - gf(f + s) \\ \frac{ds}{dt} &= r - gs(f + s)\end{aligned}$$

where $f(t)$ and $s(t)$ are the numbers of fertile and sterile individuals at time t , r is the rate of release of sterile insects, a is a positive birth rate constant, and g is a positive death rate constant.

- Show that, when no sterile insects are being released (that is, where $r = 0$), there is a locally stable equilibrium where $\hat{f} = a/g$ and $\hat{s} = 0$.
- Show that, when sterile insects are being released (that is, where $r > 0$), there is a locally stable equilibrium where $\hat{f} = 0$ and $\hat{s} = \sqrt{r/g}$.

Source: Adapted from H. Barclay et al., “Effects of Sterile Insect Releases on a Population under Predation or Parasitism,” *Researches on Population Ecology* 22 (1980): 136–146.

CASE STUDY 2d Hosts, Parasites, and Time-Travel



In this part of the case study we will take the formulation of the mathematical model for the antagonistic interactions between *Daphnia* and its parasite from Case Study 2c on page 484 and simplify it by linearization near one of its equilibrium points. We will then obtain an explicit solution for the frequency of the host and parasite genotypes as functions of time.

The analysis starts with equations

$$\begin{aligned}(1) \quad \frac{dq}{dt} &= s_q q(1 - q)(1 - 2p) \\ \frac{dp}{dt} &= s_p p(1 - p)(2q - 1)\end{aligned}$$

that were obtained in Case Study 2c. Recall that there are two possible host genotypes (A and a) and two possible parasite genotypes (B and b). Parasites of type B can infect only hosts of type A, while parasites of type b can infect only hosts of type a. In Equations 1, q is the frequency of the type A host and p is the frequency of the type B parasite. The constant s_q represents the reduction in reproductive output of a host due to infection, and s_p is the reduction in reproductive output of a parasite if it is unable to infect a host. In

Case Study 2c we found that $q = \frac{1}{2}$ and $p = \frac{1}{2}$ is an equilibrium of this system of differential equations.

Let's define $\varepsilon_q(t) = q(t) - \frac{1}{2}$ and $\varepsilon_p(t) = p(t) - \frac{1}{2}$ to be the deviations of q and p from these equilibrium values, respectively.

1. Linearize Equations 1 near the equilibrium $q = \frac{1}{2}, p = \frac{1}{2}$ to show that ε_q and ε_p satisfy the differential equations

$$(2) \quad \begin{aligned} \frac{d\varepsilon_q}{dt} &= -\frac{s_q}{2} \varepsilon_p \\ \frac{d\varepsilon_p}{dt} &= \frac{s_p}{2} \varepsilon_q \end{aligned}$$

2. Show that the solution to system (2), with initial conditions $\varepsilon_q(0)$ and $\varepsilon_p(0)$, is

$$\begin{aligned} \varepsilon_q(t) &= \varepsilon_q(0) \cos\left(\frac{1}{2} \sqrt{s_q s_p} t\right) - \varepsilon_p(0) \sqrt{\frac{s_q}{s_p}} \sin\left(\frac{1}{2} \sqrt{s_q s_p} t\right) \\ \varepsilon_p(t) &= \varepsilon_p(0) \cos\left(\frac{1}{2} \sqrt{s_q s_p} t\right) + \varepsilon_q(0) \sqrt{\frac{s_p}{s_q}} \sin\left(\frac{1}{2} \sqrt{s_q s_p} t\right) \end{aligned}$$

3. A useful trigonometric identity is

$$a \cos(ct) + b \sin(ct) = M \cos(ct - \phi)$$

where $M = \sqrt{a^2 + b^2}$ and ϕ is the angle between 0 and 2π whose cosine and sine satisfy the equations $\cos \phi = a/M$ and $\sin \phi = b/M$ (see Figure 1).

Note that if $a > 0$ and $b > 0$ (so that we are in the first quadrant), then $\phi = \tan^{-1}(b/a)$. More generally, however, ϕ is *not* given by the principal branch of $\tan^{-1}$. Instead, if $a < 0$ (second or third quadrant), then $\phi = \pi + \tan^{-1}(b/a)$, whereas if $a < 0$ and $b < 0$ (fourth quadrant), then $\phi = 2\pi + \tan^{-1}(b/a)$. Use the identity to show that the solutions in Problem 2 can be written as

$$\begin{aligned} \varepsilon_q(t) &= M_q \cos(ct - \phi_q) \\ \varepsilon_p(t) &= M_p \cos(ct - \phi_p) \end{aligned}$$

where $c = \frac{1}{2} \sqrt{s_q s_p}$,

$$\begin{aligned} M_q &= \sqrt{\varepsilon_q(0)^2 + \varepsilon_p(0)^2 \frac{s_q}{s_p}} \\ M_p &= \sqrt{\varepsilon_p(0)^2 + \varepsilon_q(0)^2 \frac{s_p}{s_q}} \end{aligned}$$

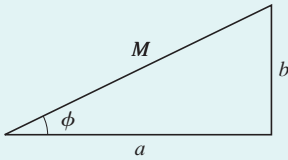


FIGURE 1

and ϕ_q and ϕ_p are given by

$$\phi_q = \begin{cases} \tan^{-1}\left(-\frac{\varepsilon_p(0)\sqrt{s_q}}{\varepsilon_q(0)\sqrt{s_p}}\right) & \text{if } \varepsilon_q(0) > 0, \varepsilon_p(0) < 0 \\ \pi + \tan^{-1}\left(-\frac{\varepsilon_p(0)\sqrt{s_q}}{\varepsilon_q(0)\sqrt{s_p}}\right) & \text{if } \varepsilon_q(0) < 0 \\ 2\pi + \tan^{-1}\left(-\frac{\varepsilon_p(0)\sqrt{s_q}}{\varepsilon_q(0)\sqrt{s_p}}\right) & \text{if } \varepsilon_q(0) > 0, \varepsilon_p(0) > 0 \end{cases}$$

and

$$\phi_p = \begin{cases} \tan^{-1}\left(-\frac{\varepsilon_q(0)\sqrt{s_p}}{\varepsilon_p(0)\sqrt{s_q}}\right) & \text{if } \varepsilon_p(0) > 0, \varepsilon_q(0) > 0 \\ \pi + \tan^{-1}\left(\frac{\varepsilon_q(0)\sqrt{s_p}}{\varepsilon_p(0)\sqrt{s_q}}\right) & \text{if } \varepsilon_p(0) < 0 \\ 2\pi + \tan^{-1}\left(\frac{\varepsilon_q(0)\sqrt{s_p}}{\varepsilon_p(0)\sqrt{s_q}}\right) & \text{if } \varepsilon_p(0) > 0, \varepsilon_q(0) < 0 \end{cases}$$

Using the definitions of $\varepsilon_q(t)$ and $\varepsilon_p(t)$, we can see that the frequencies $q(t)$ and $p(t)$, as functions of time, are given by the equations

$$\begin{aligned} q(t) &= \frac{1}{2} + M_q \cos(ct - \phi_q) \\ p(t) &= \frac{1}{2} + M_p \cos(ct - \phi_p) \end{aligned} \tag{3}$$

4. Describe, qualitatively, the predicted behavior of q and p from Equations 3.
5. How do the constants M_q and M_p affect the behavior? How do the constants ϕ_q and ϕ_p affect the behavior? How does the constant c affect the behavior?
6. Use your answers to Problem 5 to explain how the constants s_q and s_p affect the frequency of type A hosts and type B parasites over time. Can you provide a biological explanation for your answer?

The properties of Equations 3, and the predictions that can be obtained from them in terms of experimental data, are explored in Case Study 2a and 2b.

- A** Intervals, Inequalities and Absolute Values
- B** Coordinate Geometry
- C** Trigonometry
- D** Precise Definitions of Limits
- E** A Few Proofs
- F** Sigma Notation
- G** Complex Numbers

A Intervals, Inequalities, and Absolute Values



FIGURE 1
Open interval (a, b)



FIGURE 2
Closed interval $[a, b]$

Certain sets of real numbers, called **intervals**, occur frequently in calculus and correspond geometrically to line segments. For example, if $a < b$, the **open interval** from a to b consists of all numbers between a and b and is denoted by the symbol (a, b) . Using set-builder notation, we can write

$$(a, b) = \{x \mid a < x < b\}$$

Notice that the endpoints of the interval—namely, a and b —are excluded. This is indicated by the round brackets $()$ and by the open dots in Figure 1. The **closed interval** from a to b is the set

$$[a, b] = \{x \mid a \leq x \leq b\}$$

Here the endpoints of the interval are included. This is indicated by the square brackets $[\]$ and by the solid dots in Figure 2. It is also possible to include only one endpoint in an interval, as shown in Table 1.

(1) Table of Intervals

Table 1 lists the nine possible types of intervals. When these intervals are discussed, it is always assumed that $a < b$.

Notation	Set description	Picture
(a, b)	$\{x \mid a < x < b\}$	
$[a, b]$	$\{x \mid a \leq x \leq b\}$	
$[a, b)$	$\{x \mid a \leq x < b\}$	
$(a, b]$	$\{x \mid a < x \leq b\}$	
(a, ∞)	$\{x \mid x > a\}$	
$[a, \infty)$	$\{x \mid x \geq a\}$	
$(-\infty, b)$	$\{x \mid x < b\}$	
$(-\infty, b]$	$\{x \mid x \leq b\}$	
$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	

We also need to consider infinite intervals such as

$$(a, \infty) = \{x \mid x > a\}$$

This does not mean that ∞ (“infinity”) is a number. The notation (a, ∞) stands for the set of all numbers that are greater than a , so the symbol ∞ simply indicates that the interval extends indefinitely far in the positive direction.

■ Inequalities

When working with inequalities, note the following rules.

Rules for Inequalities

1. If $a < b$, then $a + c < b + c$.
2. If $a < b$ and $c < d$, then $a + c < b + d$.
3. If $a < b$ and $c > 0$, then $ac < bc$.
4. If $a < b$ and $c < 0$, then $ac > bc$.
5. If $0 < a < b$, then $1/a > 1/b$.

Rule 1 says that we can add any number to both sides of an inequality, and Rule 2 says that two inequalities can be added. However, we have to be careful with multiplication. Rule 3 says that we can multiply both sides of an inequality by a *positive number*, but Rule 4 says that **if we multiply both sides of an inequality by a negative number, then we reverse the direction of the inequality**. For example, if we take the inequality $3 < 5$ and multiply by 2, we get $6 < 10$, but if we multiply by -2 , we get $-6 > -10$. Finally, Rule 5 says that if we take reciprocals, then we reverse the direction of an inequality (provided the numbers are positive).

EXAMPLE 1 | Solve the inequality $1 + x < 7x + 5$.

SOLUTION The given inequality is satisfied by some values of x but not by others. To *solve* an inequality means to determine the set of numbers x for which the inequality is true. This is called the *solution set*.

First we subtract 1 from each side of the inequality (using Rule 1 with $c = -1$):

$$x < 7x + 4$$

Then we subtract $7x$ from both sides (Rule 1 with $c = -7x$):

$$-6x < 4$$

Now we divide both sides by -6 (Rule 4 with $c = -\frac{1}{6}$):

$$x > -\frac{4}{6} = -\frac{2}{3}$$

These steps can all be reversed, so the solution set consists of all numbers greater than $-\frac{2}{3}$. In other words, the solution of the inequality is the interval $(-\frac{2}{3}, \infty)$. ■

EXAMPLE 2 | Solve the inequality $x^2 - 5x + 6 \leq 0$.

SOLUTION First we factor the left side:

$$(x - 2)(x - 3) \leq 0$$

We know that the corresponding equation $(x - 2)(x - 3) = 0$ has the solutions 2 and 3. The numbers 2 and 3 divide the real line into three intervals:

$$(-\infty, 2) \quad (2, 3) \quad (3, \infty)$$

On each of these intervals we determine the signs of the factors. For instance,

$$x \in (-\infty, 2) \Rightarrow x < 2 \Rightarrow x - 2 < 0$$

Then we record these signs in the following chart:

Interval	$x - 2$	$x - 3$	$(x - 2)(x - 3)$
$x < 2$	−	−	+
$2 < x < 3$	+	−	−
$x > 3$	+	+	+

Another method for obtaining the information in the chart is to use *test values*. For instance, if we use the test value $x = 1$ for the interval $(-\infty, 2)$, then substitution in $x^2 - 5x + 6$ gives

$$1^2 - 5(1) + 6 = 2$$

A visual method for solving Example 2 is to use a graphing device to graph the parabola $y = x^2 - 5x + 6$ (as in Figure 3) and observe that the curve lies on or below the x -axis when $2 \leq x \leq 3$.

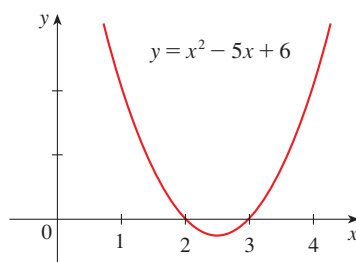


FIGURE 3

The polynomial $x^2 - 5x + 6$ doesn't change sign inside any of the three intervals, so we conclude that it is positive on $(-\infty, 2)$.

Then we read from the chart that $(x - 2)(x - 3)$ is negative when $2 < x < 3$. Thus the solution of the inequality $(x - 2)(x - 3) \leq 0$ is

$$\{x \mid 2 \leq x \leq 3\} = [2, 3]$$

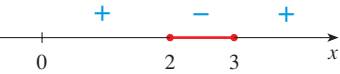


FIGURE 4

Notice that we have included the endpoints 2 and 3 because we are looking for values of x such that the product is either negative or zero. The solution is illustrated in Figure 4.

EXAMPLE 3 | Solve $x^3 + 3x^2 > 4x$.

SOLUTION First we take all nonzero terms to one side of the inequality sign and factor the resulting expression:

$$x^3 + 3x^2 - 4x > 0 \quad \text{or} \quad x(x - 1)(x + 4) > 0$$

As in Example 2 we solve the corresponding equation $x(x - 1)(x + 4) = 0$ and use the solutions $x = -4$, $x = 0$, and $x = 1$ to divide the real line into four intervals $(-\infty, -4)$, $(-4, 0)$, $(0, 1)$, and $(1, \infty)$. On each interval the product keeps a constant sign as shown in the following chart:

Interval	x	$x - 1$	$x + 4$	$x(x - 1)(x + 4)$
$x < -4$	—	—	—	—
$-4 < x < 0$	—	—	+	+
$0 < x < 1$	+	—	+	—
$x > 1$	+	+	+	+

Then we read from the chart that the solution set is

$$\{x \mid -4 < x < 0 \text{ or } x > 1\} = (-4, 0) \cup (1, \infty)$$



FIGURE 5

The solution is illustrated in Figure 5.

Absolute Value

The **absolute value** of a number a , denoted by $|a|$, is the distance from a to 0 on the real number line. Distances are always positive or 0, so we have

$$|a| \geq 0 \quad \text{for every number } a$$

For example,

$$|3| = 3 \quad |-3| = 3 \quad |0| = 0 \quad |\sqrt{2} - 1| = \sqrt{2} - 1 \quad |3 - \pi| = \pi - 3$$

In general, we have

Remember that if a is negative, then $-a$ is positive.

(2)

$|a| = a \quad \text{if } a \geq 0$
 $|a| = -a \quad \text{if } a < 0$

EXAMPLE 4 | Express $|3x - 2|$ without using the absolute-value symbol.

SOLUTION

$$\begin{aligned} |3x - 2| &= \begin{cases} 3x - 2 & \text{if } 3x - 2 \geq 0 \\ -(3x - 2) & \text{if } 3x - 2 < 0 \end{cases} \\ &= \begin{cases} 3x - 2 & \text{if } x \geq \frac{2}{3} \\ 2 - 3x & \text{if } x < \frac{2}{3} \end{cases} \end{aligned}$$

Recall that the symbol $\sqrt{}$ means “the positive square root of.” Thus $\sqrt{r} = s$ means $s^2 = r$ and $s \geq 0$. Therefore the equation $\sqrt{a^2} = a$ is not always true. It is true only when $a \geq 0$. If $a < 0$, then $-a > 0$, so we have $\sqrt{a^2} = -a$. In view of (2), we then have the equation

(3)

$$\sqrt{a^2} = |a|$$

which is true for all values of a .

Hints for the proofs of the following properties are given in the exercises.

Properties of Absolute Values Suppose a and b are any real numbers and n is an integer. Then

1. $|ab| = |a||b|$
2. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|} \quad (b \neq 0)$
3. $|a^n| = |a|^n$

For solving equations or inequalities involving absolute values, it's often very helpful to use the following statements.

Suppose $a > 0$. Then

4. $|x| = a$ if and only if $x = \pm a$
5. $|x| < a$ if and only if $-a < x < a$
6. $|x| > a$ if and only if $x > a$ or $x < -a$

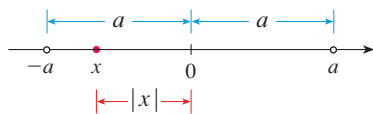


FIGURE 6

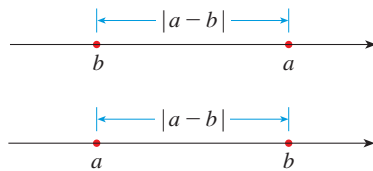


FIGURE 7

Length of a line segment = $|a - b|$

For instance, the inequality $|x| < a$ says that the distance from x to the origin is less than a , and you can see from Figure 6 that this is true if and only if x lies between $-a$ and a .

If a and b are any real numbers, then the distance between a and b is the absolute value of the difference, namely, $|a - b|$, which is also equal to $|b - a|$. (See Figure 7.)

EXAMPLE 5 | Solve $|2x - 5| = 3$.

SOLUTION By Property 4 of absolute values, $|2x - 5| = 3$ is equivalent to

$$2x - 5 = 3 \quad \text{or} \quad 2x - 5 = -3$$

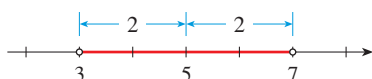
So $2x = 8$ or $2x = 2$. Thus $x = 4$ or $x = 1$.

EXAMPLE 6 | Solve $|x - 5| < 2$.**SOLUTION 1** By Property 5 of absolute values, $|x - 5| < 2$ is equivalent to

$$-2 < x - 5 < 2$$

Therefore, adding 5 to each side, we have

$$3 < x < 7$$

**FIGURE 8**and the solution set is the open interval $(3, 7)$.**SOLUTION 2** Geometrically, the solution set consists of all numbers x whose distance from 5 is less than 2. From Figure 8 we see that this is the interval $(3, 7)$. ■**EXAMPLE 7** | Solve $|3x + 2| \geq 4$.**SOLUTION** By Properties 4 and 6 of absolute values, $|3x + 2| \geq 4$ is equivalent to

$$3x + 2 \geq 4 \quad \text{or} \quad 3x + 2 \leq -4$$

In the first case $3x \geq 2$, which gives $x \geq \frac{2}{3}$. In the second case $3x \leq -6$, which gives $x \leq -2$. So the solution set is

$$\{x \mid x \leq -2 \text{ or } x \geq \frac{2}{3}\} = (-\infty, -2] \cup [\frac{2}{3}, \infty)$$

EXERCISES A**1–10** Rewrite the expression without using the absolute-value symbol.

- | | |
|-------------------------|-------------------------|
| 1. $ 5 - 23 $ | 2. $ \pi - 2 $ |
| 3. $ \sqrt{5} - 5 $ | 4. $ -2 - -3 $ |
| 5. $ x - 2 $ if $x < 2$ | 6. $ x - 2 $ if $x > 2$ |
| 7. $ x + 1 $ | 8. $ 2x - 1 $ |
| 9. $ x^2 + 1 $ | 10. $ 1 - 2x^2 $ |

11–26 Solve the inequality in terms of intervals and illustrate the solution set on the real number line.

- | | |
|------------------------------------|-------------------------------|
| 11. $2x + 7 > 3$ | 12. $4 - 3x \geq 6$ |
| 13. $1 - x \leq 2$ | 14. $1 + 5x > 5 - 3x$ |
| 15. $0 \leq 1 - x < 1$ | 16. $1 < 3x + 4 \leq 16$ |
| 17. $(x - 1)(x - 2) > 0$ | 18. $x^2 < 2x + 8$ |
| 19. $x^2 < 3$ | 20. $x^2 \geq 5$ |
| 21. $x^3 - x^2 \leq 0$ | |
| 22. $(x + 1)(x - 2)(x + 3) \geq 0$ | |
| 23. $x^3 > x$ | 24. $x^3 + 3x < 4x^2$ |
| 25. $\frac{1}{x} < 4$ | 26. $-3 < \frac{1}{x} \leq 1$ |

27. The relationship between the Celsius and Fahrenheit temperature scales is given by $C = \frac{5}{9}(F - 32)$, where C is the temperature in degrees Celsius and F is the temperature in degrees Fahrenheit. What interval on the Celsius scale corresponds to the temperature range $50 \leq F \leq 95$?
28. Use the relationship between C and F given in Exercise 27 to find the interval on the Fahrenheit scale corresponding to the temperature range $20 \leq C \leq 30$.
29. As dry air moves upward, it expands and in so doing cools at a rate of about 1°C for each 100-m rise, up to about 12 km.
- (a) If the ground temperature is 20°C , write a formula for the temperature at height h .
- (b) What range of temperature can be expected if a plane takes off and reaches a maximum height of 5 km?
30. If a ball is thrown upward from the top of a building 128 ft high with an initial velocity of 16 ft/s, then the height h above the ground t seconds later will be

$$h = 128 + 16t - 16t^2$$

During what time interval will the ball be at least 32 ft above the ground?

31–32 Solve the equation for x .

31. $|x + 3| = |2x + 1|$ 32. $|3x + 5| = 1$

33–40 Solve the inequality.

33. $|x| < 3$

34. $|x| \geq 3$

35. $|x - 4| < 1$

36. $|x - 6| < 0.1$

37. $|x + 5| \geq 2$

38. $|x + 1| \geq 3$

39. $|2x - 3| \leq 0.4$

40. $|5x - 2| < 6$

41. Solve the inequality $a(bx - c) \geq bc$ for x , assuming that a , b , and c are positive constants.

42. Solve the inequality $ax + b < c$ for x , assuming that a , b , and c are negative constants.

43. Prove that $|ab| = |a||b|$. [Hint: Use Equation 3.]

44. Show that if $0 < a < b$, then $a^2 < b^2$.

B | Coordinate Geometry

The points in a plane can be identified with ordered pairs of real numbers. We start by drawing two perpendicular coordinate lines that intersect at the origin O on each line. Usually one line is horizontal with positive direction to the right and is called the x -axis; the other line is vertical with positive direction upward and is called the y -axis.

Any point P in the plane can be located by a unique ordered pair of numbers as follows: Draw lines through P perpendicular to the x - and y -axes. These lines intersect the axes in points with coordinates a and b as shown in Figure 1. Then the point P is assigned the ordered pair (a, b) . The first number a is called the **x -coordinate** of P ; the second number b is called the **y -coordinate** of P . We say that P is the point with coordinates (a, b) , and we denote the point by the symbol $P(a, b)$. Several points are labeled with their coordinates in Figure 2.

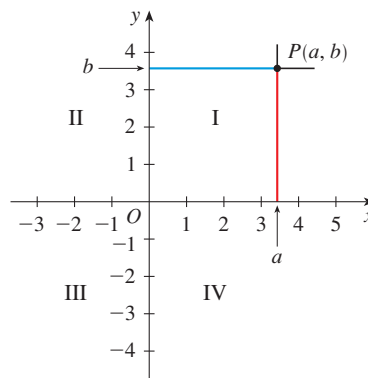


FIGURE 1

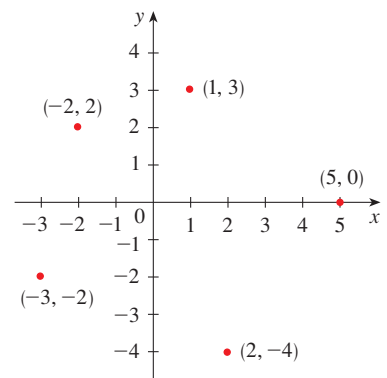


FIGURE 2

By reversing the preceding process we can start with an ordered pair (a, b) and arrive at the corresponding point P . Often we identify the point P with the ordered pair (a, b) and refer to “the point (a, b) .” [Although the notation used for an open interval (a, b) is the same as the notation used for a point (a, b) , you will be able to tell from the context which meaning is intended.]

This coordinate system is called the **rectangular coordinate system** or the **Cartesian coordinate system** in honor of the French mathematician René Descartes (1596–1650), even though another Frenchman, Pierre Fermat (1601–1665), invented the principles of analytic geometry at about the same time as Descartes. The plane supplied with this coordinate system is called the **coordinate plane** or the **Cartesian plane** and is denoted by $\mathbb{R}^2$.

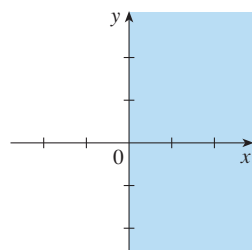
The x - and y -axes are called the **coordinate axes** and divide the Cartesian plane into four quadrants, which are labeled I, II, III, and IV in Figure 1. Notice that the first quadrant consists of those points whose x - and y -coordinates are both positive.

EXAMPLE 1 | Describe and sketch the regions given by the following sets.

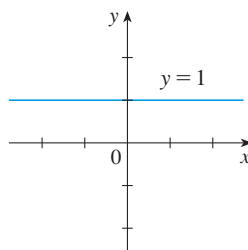
- (a) $\{(x, y) \mid x \geq 0\}$ (b) $\{(x, y) \mid y = 1\}$ (c) $\{(x, y) \mid |y| < 1\}$

SOLUTION

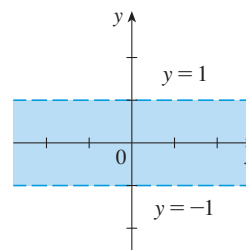
(a) The points whose x -coordinates are 0 or positive lie on the y -axis or to the right of it as indicated by the shaded region in Figure 3(a).



(a) $x \geq 0$



(b) $y = 1$



(c) $|y| < 1$

FIGURE 3

(b) The set of all points with y -coordinate 1 is a horizontal line one unit above the x -axis [see Figure 3(b)].

(c) Recall from Appendix A that

$$|y| < 1 \quad \text{if and only if} \quad -1 < y < 1$$

The given region consists of those points in the plane whose y -coordinates lie between -1 and 1 . Thus the region consists of all points that lie between (but not on) the horizontal lines $y = 1$ and $y = -1$. [These lines are shown as dashed lines in Figure 3(c) to indicate that the points on these lines don't lie in the set.] ■

Recall from Appendix A that the distance between points a and b on a number line is $|a - b| = |b - a|$. Thus the distance between points $P_1(x_1, y_1)$ and $P_3(x_2, y_1)$ on a horizontal line must be $|x_2 - x_1|$ and the distance between $P_2(x_2, y_2)$ and $P_3(x_2, y_1)$ on a vertical line must be $|y_2 - y_1|$. (See Figure 4.)

To find the distance $|P_1P_2|$ between any two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$, we note that triangle $P_1P_2P_3$ in Figure 4 is a right triangle, and so by the Pythagorean Theorem we have

$$\begin{aligned} |P_1P_2| &= \sqrt{|P_1P_3|^2 + |P_2P_3|^2} = \sqrt{|x_2 - x_1|^2 + |y_2 - y_1|^2} \\ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \end{aligned}$$

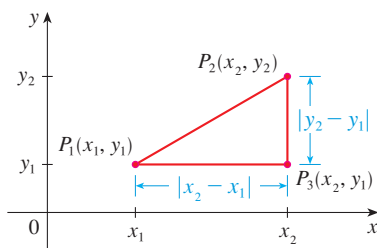


FIGURE 4

Distance Formula The distance between the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For instance, the distance between $(1, -2)$ and $(5, 3)$

$$\sqrt{(5 - 1)^2 + [3 - (-2)]^2} = \sqrt{4^2 + 5^2} = \sqrt{41}$$

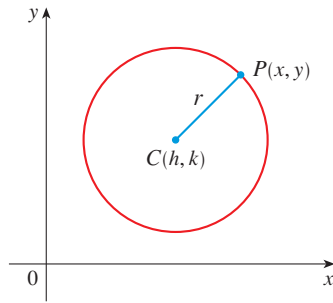


FIGURE 5

Circles

An **equation of a curve** is an equation satisfied by the coordinates of the points on the curve and by no other points. Let's use the distance formula to find the equation of a circle with radius r and center (h, k) . By definition, the circle is the set of all points $P(x, y)$ whose distance from the center $C(h, k)$ is r . (See Figure 5.) Thus P is on the circle if and only if $|PC| = r$. From the distance formula, we have

$$\sqrt{(x - h)^2 + (y - k)^2} = r$$

or equivalently, squaring both sides, we get

$$(x - h)^2 + (y - k)^2 = r^2$$

This is the desired equation.

Equation of a Circle An equation of the circle with center (h, k) and radius r is

$$(x - h)^2 + (y - k)^2 = r^2$$

In particular, if the center is the origin $(0, 0)$, the equation is

$$x^2 + y^2 = r^2$$

For instance an equation of the circle with radius 3 and center $(2, -5)$ is

$$(x - 2)^2 + (y + 5)^2 = 9$$

EXAMPLE 2 | Sketch the graph of the equation $x^2 + y^2 + 2x - 6y + 7 = 0$ by first showing that it represents a circle and then finding its center and radius.

SOLUTION We first group the x -terms and y -terms as follows:

$$(x^2 + 2x) + (y^2 - 6y) = -7$$

Then we complete the square within each grouping, adding the appropriate constants (the squares of half the coefficients of x and y) to both sides of the equation:

$$(x^2 + 2x + 1) + (y^2 - 6y + 9) = -7 + 1 + 9$$

or

$$(x + 1)^2 + (y - 3)^2 = 3$$

Comparing this equation with the standard equation of a circle, we see that $h = -1$, $k = 3$, and $r = \sqrt{3}$, so the given equation represents a circle with center $(-1, 3)$ and radius $\sqrt{3}$. It is sketched in Figure 6.

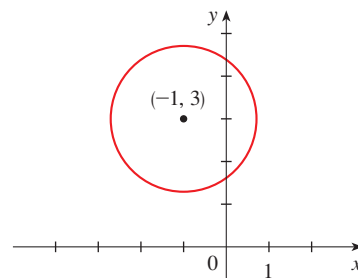


FIGURE 6
 $x^2 + y^2 + 2x - 6y + 7 = 0$

Lines

To find the equation of a line L we use its *slope*, which is a measure of the steepness of the line.

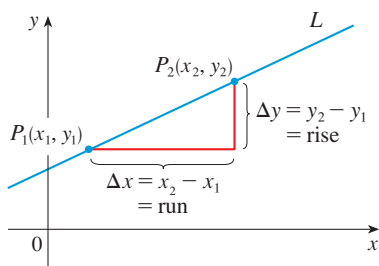


FIGURE 7

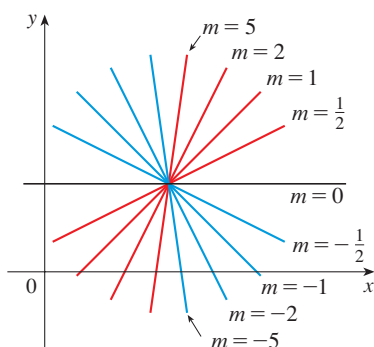


FIGURE 8

Definition The **slope** of a nonvertical line that passes through the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

The slope of a vertical line is not defined.

Thus the slope of a line is the ratio of the change in y , Δy , to the change in x , Δx . (See Figure 7.) The slope is therefore the rate of change of y with respect to x . The fact that the line is straight means that the rate of change is constant.

Figure 8 shows several lines labeled with their slopes. Notice that lines with positive slope slant upward to the right, whereas lines with negative slope slant downward to the right. Notice also that the steepest lines are the ones for which the absolute value of the slope is largest, and a horizontal line has slope 0.

Now let's find an equation of the line that passes through a given point $P_1(x_1, y_1)$ and has slope m . A point $P(x, y)$ with $x \neq x_1$ lies on this line if and only if the slope of the line through P_1 and P is equal to m ; that is,

$$\frac{y - y_1}{x - x_1} = m$$

This equation can be rewritten in the form

$$y - y_1 = m(x - x_1)$$

and we observe that this equation is also satisfied when $x = x_1$ and $y = y_1$. Therefore it is an equation of the given line.

Point-Slope Form of the Equation of a Line An equation of the line passing through the point $P_1(x_1, y_1)$ and having slope m is

$$y - y_1 = m(x - x_1)$$

EXAMPLE 3 Find an equation of the line through the points $(-1, 2)$ and $(3, -4)$.

SOLUTION The slope of the line is

$$m = \frac{-4 - 2}{3 - (-1)} = -\frac{3}{2}$$

Using the point-slope form with $x_1 = -1$ and $y_1 = 2$, we obtain

$$y - 2 = -\frac{3}{2}(x + 1)$$

which simplifies to

$$3x + 2y = 1$$

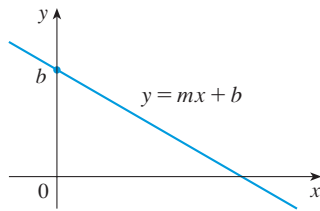


FIGURE 9

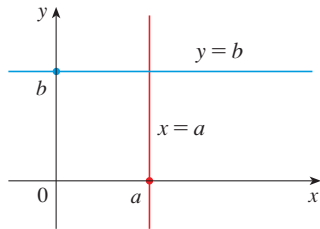


FIGURE 10

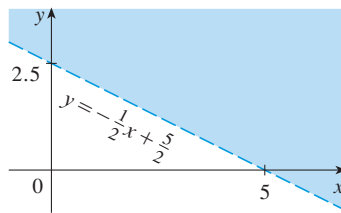


FIGURE 11

Suppose a nonvertical line has slope m and y -intercept b . (See Figure 9.) This means it intersects the y -axis at the point $(0, b)$, so the point-slope form of the equation of the line, with $x_1 = 0$ and $y_1 = b$, becomes

$$y - b = m(x - 0)$$

This simplifies as follows.

Slope-Intercept Form of the Equation of a Line An equation of the line with slope m and y -intercept b is

$$y = mx + b$$

In particular, if a line is horizontal, its slope is $m = 0$, so its equation is $y = b$, where b is the y -intercept (see Figure 10). A vertical line does not have a slope, but we can write its equation as $x = a$, where a is the x -intercept, because the x -coordinate of every point on the line is a .

EXAMPLE 4 | Graph the inequality $x + 2y > 5$.

SOLUTION We are asked to sketch the graph of the set $\{(x, y) \mid x + 2y > 5\}$ and we begin by solving the inequality for y :

$$\begin{aligned} x + 2y &> 5 \\ 2y &> -x + 5 \\ y &> -\frac{1}{2}x + \frac{5}{2} \end{aligned}$$

Compare this inequality with the equation $y = -\frac{1}{2}x + \frac{5}{2}$, which represents a line with slope $-\frac{1}{2}$ and y -intercept $\frac{5}{2}$. We see that the given graph consists of points whose y -coordinates are *larger* than those on the line $y = -\frac{1}{2}x + \frac{5}{2}$. Thus the graph is the region that lies *above* the line, as illustrated in Figure 11.

Parallel and Perpendicular Lines

Slopes can be used to show that lines are parallel or perpendicular. The following facts are proved, for instance, in *Precalculus: Mathematics for Calculus*, Sixth Edition, by Stewart, Redlin, and Watson (Belmont, CA, 2012).

Parallel and Perpendicular Lines

- Two nonvertical lines are parallel if and only if they have the same slope.
- Two lines with slopes m_1 and m_2 are perpendicular if and only if $m_1 m_2 = -1$; that is, their slopes are negative reciprocals:

$$m_2 = -\frac{1}{m_1}$$

EXAMPLE 5 | Find an equation of the line through the point $(5, 2)$ that is parallel to the line $4x + 6y + 5 = 0$.

SOLUTION The given line can be written in the form

$$y = -\frac{2}{3}x - \frac{5}{6}$$

which is in slope-intercept form with $m = -\frac{2}{3}$. Parallel lines have the same slope, so the required line has slope $-\frac{2}{3}$ and its equation in point-slope form is

$$y - 2 = -\frac{2}{3}(x - 5)$$

We can write this equation as $2x + 3y = 16$. ■

EXAMPLE 6 | Show that the lines $2x + 3y = 1$ and $6x - 4y - 1 = 0$ are perpendicular.

SOLUTION The equations can be written as

$$y = -\frac{2}{3}x + \frac{1}{3} \quad \text{and} \quad y = \frac{3}{2}x - \frac{1}{4}$$

from which we see that the slopes are

$$m_1 = -\frac{2}{3} \quad \text{and} \quad m_2 = \frac{3}{2}$$

Since $m_1 m_2 = -1$, the lines are perpendicular. ■

Conic Sections

Here we review the geometric definitions of parabolas, ellipses, and hyperbolas and their standard equations. They are called **conic sections**, or **conics**, because they result from intersecting a cone with a plane as shown in Figure 12.

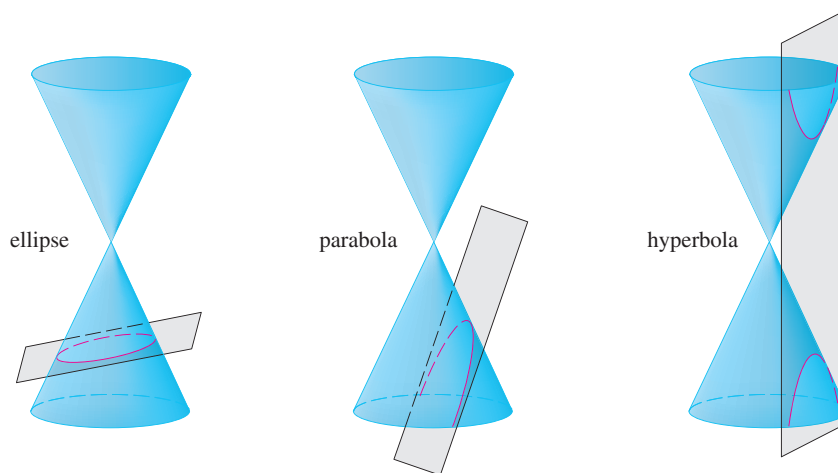


FIGURE 12
Conics

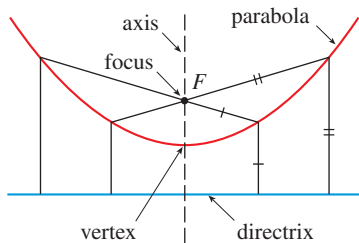


FIGURE 13

Parabolas

A **parabola** is the set of points in a plane that are equidistant from a fixed point F (called the **focus**) and a fixed line (called the **directrix**). This definition is illustrated by Figure 13. Notice that the point halfway between the focus and the directrix lies on the parabola; it is called the **vertex**. The line through the focus perpendicular to the directrix is called the **axis** of the parabola.

In the 16th century Galileo showed that the path of a projectile that is shot into the air at an angle to the ground is a parabola. Since then, parabolic shapes have been used in designing automobile headlights, reflecting telescopes, and suspension bridges.

We obtain a particularly simple equation for a parabola if we place its vertex at the origin O and its directrix parallel to the x -axis as in Figure 14. If the focus is the point

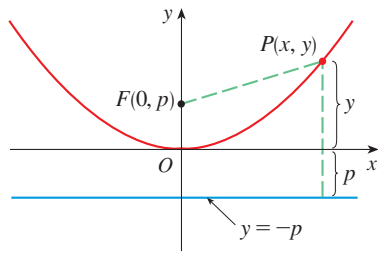


FIGURE 14

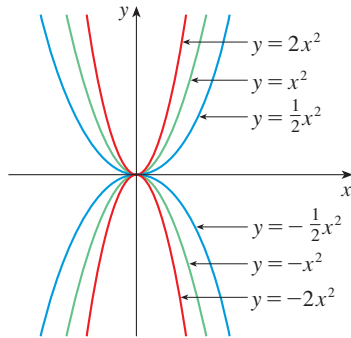


FIGURE 15

$(0, p)$, then the directrix has the equation $y = -p$ and the parabola has the equation

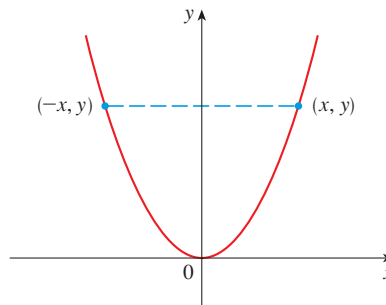
$$x^2 = 4py$$

(See Exercise 47.)

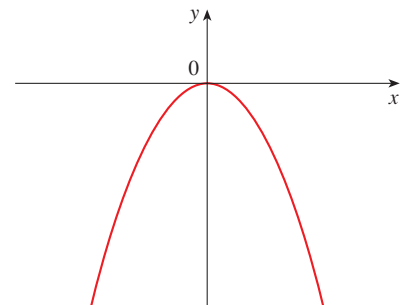
If we write $a = 1/(4p)$, then the equation of a parabola becomes

$$y = ax^2$$

Figure 15 shows the graphs of several parabolas with equations of the form $y = ax^2$ for various values of the number a . We see that the parabola $y = ax^2$ opens upward if $a > 0$ and downward if $a < 0$ (as in Figure 16). The graph is symmetric with respect to the y -axis because its equation is unchanged when x is replaced by $-x$. This corresponds to the fact that the function $f(x) = ax^2$ is an even function.

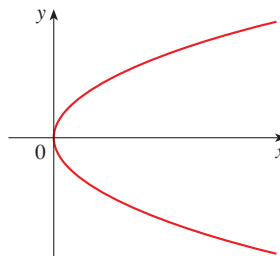


(a) $y = ax^2, a > 0$

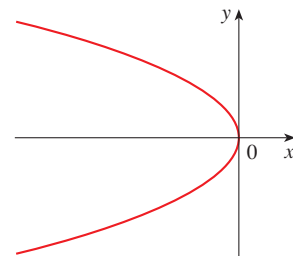


(b) $y = ax^2, a < 0$

If we interchange x and y in the equation $y = ax^2$, the result is $x = ay^2$, which also represents a parabola. (Interchanging x and y amounts to reflecting about the diagonal line $y = x$.) The parabola $x = ay^2$ opens to the right if $a > 0$ and to the left if $a < 0$. (See Figure 17.) This time the parabola is symmetric with respect to the x -axis because the equation is unchanged when y is replaced by $-y$.



(a) $x = ay^2, a > 0$



(b) $x = ay^2, a < 0$

FIGURE 17

EXAMPLE 7 | Sketch the region bounded by the parabola $x = 1 - y^2$ and the line $x + y + 1 = 0$.

SOLUTION First we find the points of intersection by solving the two equations. Substituting $x = -y - 1$ into the equation $x = 1 - y^2$, we get $-y - 1 = 1 - y^2$, which gives

$$0 = y^2 - y - 2 = (y - 2)(y + 1)$$

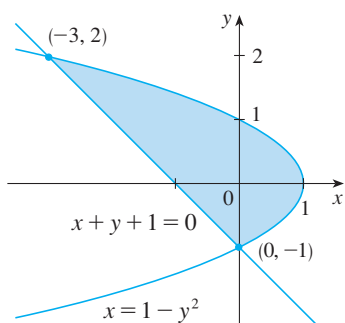


FIGURE 18

so $y = 2$ or -1 . Thus the points of intersection are $(-3, 2)$ and $(0, -1)$ and we draw the line $x + y + 1 = 0$ passing through these points.

To sketch the parabola $x = 1 - y^2$ we start with the parabola $x = -y^2$ in Figure 17(b) and shift one unit to the right. We also make sure it passes through the points $(-3, 2)$ and $(0, -1)$. The region bounded by $x = 1 - y^2$ and $x + y + 1 = 0$ means the finite region whose boundaries are these curves. It is sketched in Figure 18.

■ Ellipses

An **ellipse** is the set of points in a plane the sum of whose distances from two fixed points F_1 and F_2 is a constant (see Figure 19). These two fixed points are called the **foci** (plural of **focus**). One of Kepler's laws is that the orbits of the planets in the solar system are ellipses with the sun at one focus.

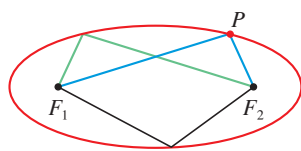


FIGURE 19

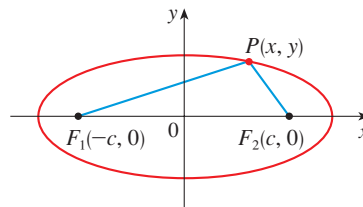


FIGURE 20

In order to obtain the simplest equation for an ellipse, we place the foci on the x -axis at the points $(-c, 0)$ and $(c, 0)$ as in Figure 20 so that the origin is halfway between the foci. If we let the sum of the distances from a point on the ellipse to the foci be $2a$, then we can write an equation of the ellipse as

(1)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where $c^2 = a^2 - b^2$. (See Exercise 49 and Figure 21.) Notice that the x -intercepts are $\pm a$, the y -intercepts are $\pm b$, the foci are $(\pm c, 0)$, and the ellipse is symmetric with respect to both axes. If the foci of an ellipse are located on the y -axis at $(0, \pm c)$, then we can find its equation by interchanging x and y in (1).

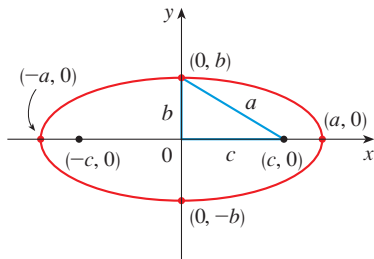


FIGURE 21

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a \geq b$$

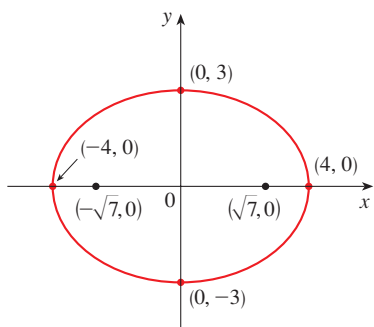


FIGURE 22

$$9x^2 + 16y^2 = 144$$

EXAMPLE 8 | Sketch the graph of $9x^2 + 16y^2 = 144$ and locate the foci.

SOLUTION Divide both sides of the equation by 144:

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

The equation is now in the standard form for an ellipse, so we have $a^2 = 16$, $b^2 = 9$, $a = 4$, and $b = 3$. The x -intercepts are ± 4 and the y -intercepts are ± 3 . Also, $c^2 = a^2 - b^2 = 7$, so $c = \sqrt{7}$ and the foci are $(\pm\sqrt{7}, 0)$. The graph is sketched in Figure 22.

Like parabolas, ellipses have an interesting reflection property that has practical consequences. If a source of light or sound is placed at one focus of a surface with elliptical cross-sections, then all the light or sound is reflected off the surface to the other focus. This principle is used in *lithotripsy*, a treatment for kidney stones. A reflector with elliptical cross-section is placed in such a way that the kidney stone is at one focus. High-

intensity sound waves generated at the other focus are reflected to the stone and destroy it without damaging surrounding tissue. The patient is spared the trauma of surgery and recovers within a few days.

Hyperbolas

A **hyperbola** is the set of all points in a plane the difference of whose distances from two fixed points F_1 and F_2 (the foci) is a constant. This definition is illustrated in Figure 23.

Notice that the definition of a hyperbola is similar to that of an ellipse; the only change is that the sum of distances has become a difference of distances. It is left as Exercise 51 to show that when the foci are on the x -axis at $(\pm c, 0)$ and the difference of distances is $|PF_1| - |PF_2| = \pm 2a$, then the equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where $c^2 = a^2 + b^2$. Notice that the x -intercepts are again $\pm a$. But if we put $x = 0$ in Equation 2 we get $y^2 = -b^2$, which is impossible, so there is no y -intercept. The hyperbola is symmetric with respect to both axes.

To analyze the hyperbola further, we look at Equation 2 and obtain

$$\frac{x^2}{a^2} = 1 + \frac{y^2}{b^2} \geq 1$$

This shows that $x^2 \geq a^2$, so $|x| = \sqrt{x^2} \geq a$. Therefore we have $x \geq a$ or $x \leq -a$. This means that the hyperbola consists of two parts, called its *branches*.

When we draw a hyperbola it is useful to first draw its *asymptotes*, which are the lines $y = (b/a)x$ and $y = -(b/a)x$ shown in Figure 24. Both branches of the hyperbola approach the asymptotes; that is, they come arbitrarily close to the asymptotes. If the foci of a hyperbola are on the y -axis, we find its equation by reversing the roles of x and y .

EXAMPLE 9 Find the foci and asymptotes of the hyperbola $9x^2 - 16y^2 = 144$ and sketch its graph.

SOLUTION If we divide both sides of the equation by 144, it becomes

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

which is of the form given in (2) with $a = 4$ and $b = 3$. Since $c^2 = 16 + 9 = 25$, the foci are $(\pm 5, 0)$. The asymptotes are the lines $y = \frac{3}{4}x$ and $y = -\frac{3}{4}x$. The graph is shown in Figure 25.

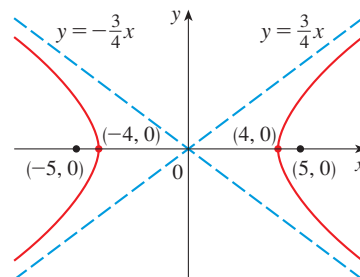


FIGURE 25
 $9x^2 - 16y^2 = 144$

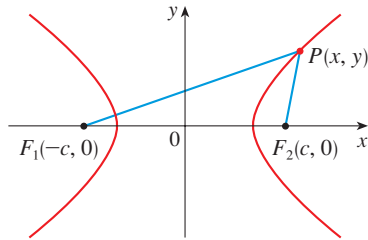


FIGURE 23
 P is on the hyperbola when
 $|PF_1| - |PF_2| = \pm 2a$.

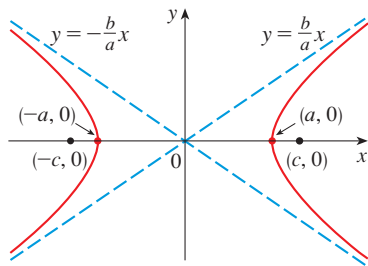


FIGURE 24
 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

EXERCISES B

1–2 Find the distance between the points.

1. $(1, 1)$, $(4, 5)$ 2. $(1, -3)$, $(5, 7)$

3–4 Find the slope of the line through P and Q .

3. $P(-3, 3)$, $Q(-1, -6)$ 4. $P(-1, -4)$, $Q(6, 0)$

5. Show that the points $(-2, 9)$, $(4, 6)$, $(1, 0)$, and $(-5, 3)$ are the vertices of a square.

6. (a) Show that the points $A(-1, 3)$, $B(3, 11)$, and $C(5, 15)$ are collinear (lie on the same line) by showing that $|AB| + |BC| = |AC|$.

(b) Use slopes to show that A , B , and C are collinear.

7–10 Sketch the graph of the equation.

7. $x = 3$ 8. $y = -2$
9. $xy = 0$ 10. $|y| = 1$

11–24 Find an equation of the line that satisfies the given conditions.

11. Through $(2, -3)$, slope 6
12. Through $(-3, -5)$, slope $-\frac{7}{2}$
13. Through $(2, 1)$ and $(1, 6)$
14. Through $(-1, -2)$ and $(4, 3)$
15. Slope 3, y -intercept -2
16. Slope $\frac{2}{5}$, y -intercept 4
17. x -intercept 1, y -intercept -3
18. x -intercept -8 , y -intercept 6
19. Through $(4, 5)$, parallel to the x -axis
20. Through $(4, 5)$, parallel to the y -axis
21. Through $(1, -6)$, parallel to the line $x + 2y = 6$
22. y -intercept 6, parallel to the line $2x + 3y + 4 = 0$
23. Through $(-1, -2)$, perpendicular to the line $2x + 5y + 8 = 0$
24. Through $(\frac{1}{2}, -\frac{2}{3})$, perpendicular to the line $4x - 8y = 1$

25–28 Find the slope and y -intercept of the line and draw its graph.

25. $x + 3y = 0$ 26. $2x - 3y + 6 = 0$
27. $3x - 4y = 12$ 28. $4x + 5y = 10$

29–36 Sketch the region in the xy -plane.

29. $\{(x, y) \mid x < 0\}$ 30. $\{(x, y) \mid x \geq 1 \text{ and } y < 3\}$
31. $\{(x, y) \mid |x| \leq 2\}$
32. $\{(x, y) \mid |x| < 3 \text{ and } |y| < 2\}$
33. $\{(x, y) \mid 0 \leq y \leq 4 \text{ and } x \leq 2\}$
34. $\{(x, y) \mid y > 2x - 1\}$
35. $\{(x, y) \mid 1 + x \leq y \leq 1 - 2x\}$
36. $\{(x, y) \mid -x \leq y < \frac{1}{2}(x + 3)\}$

37–38 Find an equation of a circle that satisfies the given conditions.

37. Center $(3, -1)$, radius 5
38. Center $(-1, 5)$, passes through $(-4, -6)$

39–40 Show that the equation represents a circle and find the center and radius.

39. $x^2 + y^2 - 4x + 10y + 13 = 0$
40. $x^2 + y^2 + 6y + 2 = 0$

41. Show that the lines $2x - y = 4$ and $6x - 2y = 10$ are not parallel and find their point of intersection.
42. Show that the lines $3x - 5y + 19 = 0$ and $10x + 6y - 50 = 0$ are perpendicular and find their point of intersection.
43. Show that the midpoint of the line segment from $P_1(x_1, y_1)$ to $P_2(x_2, y_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

44. Find the midpoint of the line segment joining the points $(1, 3)$ and $(7, 15)$.
45. Find an equation of the perpendicular bisector of the line segment joining the points $A(1, 4)$ and $B(7, -2)$.
46. (a) Show that if the x - and y -intercepts of a line are nonzero numbers a and b , then the equation of the line can be put in the form

$$\frac{x}{a} + \frac{y}{b} = 1$$

This equation is called the **two-intercept form** of an equation of a line.

- (b) Use part (a) to find an equation of the line whose x -intercept is 6 and whose y -intercept is -8 .

47. Suppose that $P(x, y)$ is any point on the parabola with focus $(0, p)$ and directrix $y = -p$. (See Figure 14.) Use the definition of a parabola to show that $x^2 = 4py$.
48. Find the focus and directrix of the parabola $y = x^2$. Illustrate with a diagram.
49. Suppose an ellipse has foci $(\pm c, 0)$ and the sum of the distances from any point $P(x, y)$ on the ellipse to the foci is $2a$. Show that the coordinates of P satisfy Equation 1.
50. Find the foci of the ellipse $x^2 + 4y^2 = 4$ and sketch its graph.
51. Use the definition of a hyperbola to derive Equation 2 for a hyperbola with foci $(\pm c, 0)$.
52. (a) Find the foci and asymptotes of the hyperbola $x^2 - y^2 = 1$ and sketch its graph.
(b) Sketch the graph of $y^2 - x^2 = 1$.
- 53–54** Sketch the region bounded by the curves.
53. $x + 4y = 8$ and $x = 2y^2 - 8$
54. $y = 4 - x^2$ and $x - 2y = 2$

C Trigonometry

Here we review the aspects of trigonometry that are used in calculus: radian measure, trigonometric functions, trigonometric identities, and inverse trigonometric functions.

■ Angles

Angles can be measured in degrees or in radians (abbreviated as rad). The angle given by a complete revolution contains 360° , which is the same as 2π rad. Therefore

(1)

$$\pi \text{ rad} = 180^\circ$$

and

(2) $1 \text{ rad} = \left(\frac{180}{\pi}\right)^\circ \approx 57.3^\circ \quad 1^\circ = \frac{\pi}{180} \text{ rad} \approx 0.017 \text{ rad}$

EXAMPLE 1

- (a) Find the radian measure of 60° . (b) Express $5\pi/4$ rad in degrees.

SOLUTION

(a) From Equation 1 or 2 we see that to convert from degrees to radians we multiply by $\pi/180$. Therefore

$$60^\circ = 60 \left(\frac{\pi}{180} \right) = \frac{\pi}{3} \text{ rad}$$

(b) To convert from radians to degrees we multiply by $180/\pi$. Thus

$$\frac{5\pi}{4} \text{ rad} = \frac{5\pi}{4} \left(\frac{180}{\pi} \right) = 225^\circ$$